Credit Card Fraud Detection Using Machine Learning Techniques

Subtitle if required

By

DEVADHARSAN SELVENDRAN

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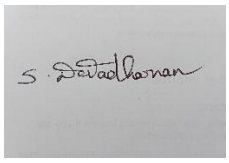
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I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

DEVADHARSAN SELVENDRAN

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Abstract

Purpose: The purpose of this research is to explore how Data Science and Machine Learning can be used for predicting obesity levels based on demographic, lifestyle and behavioural factors. The purpose of the study is to assess how predictive models of obesity can be used to aid in early identification and preventive healthcare. Additionally, the study also evaluates the performance of various machine learning algorithms and identifies the most appropriate model for precise obesity prediction, which is useful for clinical decision-making.

Methodology: This study used a quantitative research approach, with a publicly available obesity prediction dataset that included demographic data, food consumption and behavioural and physical activity attributes. Data pre-processing was applied to the dataset that were Data cleaning, Feature encoding, Feature Scaling and Train/Test Split. Four machine learning algorithms were implemented Decision Tree, Support Vector Machine, Random Forest and Deep Learning, with the baseline model being Logistic Regression. Hyperparameter tuning and cross-validation methods have been used to enhance the performance of the model and prevent overfitting. To evaluate the developed models and their predictive ability and reliability, the standard metrics used for classification such as Accuracy, Precision, Recall, F1-score, Confusion Matrix and ROC-AUC were employed.

Findings: The results showed that machine learning methods can be effectively used to predict the obesity rate with demographic and lifestyle features. Such hyperparameter tuning boosted the performance of Decision Tree and Support Vector Machine as well as the Random Forest model and each classification precision of the overall accuracy reached 97%, while the Logistic Regression model is 90% and the Deep Learning model is 92%. Cross-validation also demonstrated that the Random Forest Classifier is robust and stable and comparative analysis showed that it is the most reliable model due to its well-balanced predictive performance, excellent generalisation ability and low classification error. Weight, overweight in family history, physical activity, diet and age as high-calorie foods consumption were also identified in the study as the most influential in the prediction of obesity.

Recommendations: The authors suggest that clinical decision support systems be supplemented with machine learning models that predict obesity to enable early detection and personalised interventions and prevention in healthcare planning. Future research is recommended to involve

bigger clinical datasets, more health indicators and Explainable Artificial Intelligence techniques to enhance the transparency and potential implementation of the models in the healthcare sector.

Limitations: This study used a publicly available secondary data set, which may not accurately reflect the diversity of the clinical population. The models developed are used for a single data set and have a limited generalisability to external settings. Also, the chosen machine learning algorithms were limited, and more sophisticated ensemble- or transformer-based models could be used to enhance predictive performance.

Keywords: Obesity Prediction, Machine Learning, Data Science, Healthcare Analytics, Random Forest, Decision Tree, Support Vector Machine, Deep Learning, Predictive Healthcare, Preventive Healthcare.

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Chapter 1 Introduction

The rapid growth of online banking, e-commerce and contactless payments has made credit-card transactions faster and more convenient. But the same expansion has come with an increased number of opportunities for fraudulent activity. In financial institutions, many transactions occur on a frequent basis and involve a large volume, making manual checking process time consuming, costly and impractical for real-time decision making [1].Known fraud patterns can be identified using traditional rule-based systems, but fixed rules may not be effective if the fraud behaviour changes. Machine-learning techniques are an appealing solution, as they are able to identify patterns in previous transaction information, and determine if a new transaction is authentic or fraudulent [2].

1.1 Problem Description, Context and Motivation

Credit-card fraud is difficult to detect in real time because fraudulent payments are hidden among legitimate transactions, while criminal methods constantly change. Known activities can be detected by traditional rule-based systems, but new patterns may not be detected. In the first half of 2025, the UK experienced an estimated 1.94 million unauthorised card-fraud cases and losses of £299 million [3]. Cardholders, banks, merchants and payment providers are impacted by fraud in terms of financial losses, stalled payments, investigation expenses, and diminished overall trust. Using machine learning, patterns from past transactions can be learned and abnormal activity can be detected, reducing the size of this problem. Ensemble algorithms and Class weighting can help detect rare fraud cases with lesser alerts, and can help to aid quicker and better decision-making [1].

Research question: Which supervised machine-learning algorithm provides the most reliable credit-card fraud detection while minimising false-positive and false-negative classifications?

1.2 Objectives

1.2.1 Research aim

To develop and evaluate machine learning models to identify fraudulent credit card transactions using preprocessing, imbalance treatment and comparison.

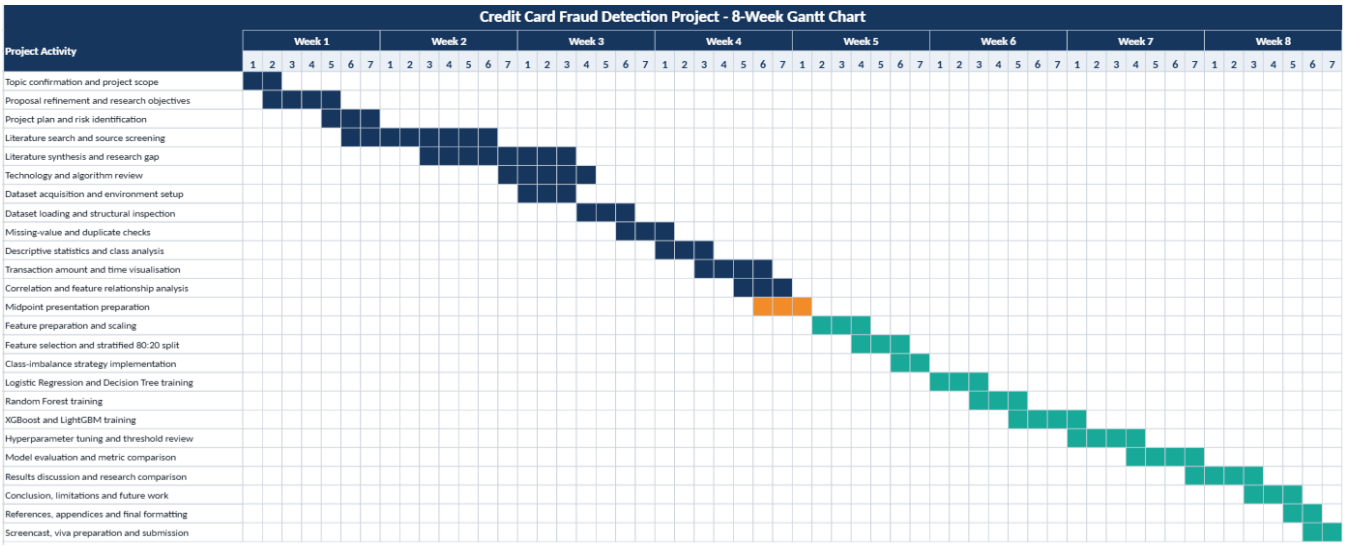
1.2.2 Research Objectives

- To analyse credit card transaction data to identify significant patterns distinguishing genuine transactions from fraudulent financial activities using preprocessing.

- To examine credit card transaction fraud challenges and factors influencing detection effectiveness across financial systems using machine learning techniques.
- To develop and compare multiple machine learning classification models for accurately detecting fraudulent credit card transactions using performance metrics.
- To evaluate classification model performance using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix for reliable fraud detection.
- To provide recommendations for improving credit card fraud detection systems using optimised machine learning models and future research directions.

1.3 Methodology

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The project adopts a quantitative machine learning approach using the Kaggle Credit Card Fraud Detection dataset. Data preprocessing includes duplicate removal, feature scaling, exploratory data analysis, stratified train-test splitting and class-weight handling for imbalance. Logistic Regression, Decision Tree, Random Forest, XGBoost and LightGBM are implemented and compared using Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix. Python, Scikit-learn and Jupyter Notebook support implementation and evaluation.

1.4 Legal, Social, Ethical and Professional Considerations

This project uses a publicly available anonymised dataset from Kaggle; therefore, no personal or identifiable information is processed, and formal ethical approval was not required. Data are used just for academic research and stored securely in accordance with good research practice. Ethical considerations include reducing false positives that inconvenience customers and false negatives

that allow fraudulent transactions to continue. Socially, improving fraud detection enhances trust in digital payment systems and reduces financial losses. Professionally, the project follows transparent reporting, reproducible methodology, accurate citation of sources and objective evaluation using appropriate performance metrics for imbalanced datasets.

1.5 Background

Credit card fraud has become a significant challenge due to the rapid growth of online banking, e-commerce and contactless payments. Financial institutions process millions of transactions daily, making manual fraud detection inefficient and costly. Traditional rule-based systems struggle to identify evolving fraud patterns, increasing the need for intelligent machine learning solutions. Supervised learning techniques can analyse historical transaction data to distinguish fraudulent from genuine activities with greater accuracy. Recent studies highlight the importance of preprocessing, imbalance handling and ensemble learning methods for improving fraud detection performance, supporting the development of reliable, scalable and practical financial security systems.

1.6 Structure of Report

Chapter 1 provides an introduction to the project, problem definition, project goals and the methodology and related legal, social, ethical and professional considerations. Literature review on fraud-detection challenges, preprocessing, machine-learning algorithms, evaluation measures and the selected technologies is reviewed in Chapter 2. The following Chapter 3 explains the experimental design, dataset preparation, exploratory analysis, preprocessing and model implementation. The results of the evaluations are presented and compared in Chapter 4. The findings are summarised in Chapter 5, suggestions for future work are made and reflections are provided on the project process.

Chapter 2 Literature – Technology Review

2.1 Literature Review

2.1.1 Introduction to Credit Card Fraud Detection

The rapid expansion of digital banking, e-commerce and contactless payment technologies has transformed global financial transactions while simultaneously increasing opportunities for credit card fraud. Manual fraud detection is impractical for financial institutions, which process millions of transactions a day, and it can have a huge impact on their balance sheets and brand reputation. According to the study conducted by [1] stated that conventional rule-driven fraud detection systems are becoming obsolete since they are driven by handcrafted rules that cannot keep up with evolving strategies. Similarly, [2] pointed out that the fraud patterns of today demand intelligent systems with the ability to learn hidden statistical relationships and [3] demonstrated the effectiveness of the use of deep learning models to identify complex nonlinear patterns of transactions. Additionally, as per the views of [4] Privacy-preserving architectures provide better fraud detection without compromising data security. However, [5] claimed that there are limitations in terms of implementing it in practice, including data imbalance, concept drift and computational complexity. As a result, recent research has been directed towards the application of intelligent machine learning tools that can learn complex fraud patterns and be able to detect fraud in a scalable and real-time manner.

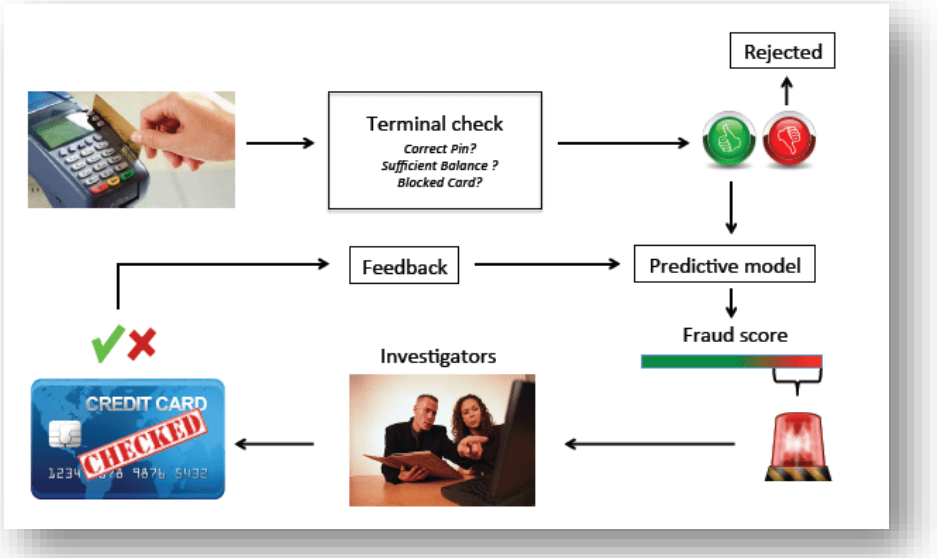


Figure 1: Credit Card Fraud Detection

(Source: [6]***)***

2.1.2 Credit Card Fraud and Fraud Detection Challenges

Credit card fraud has evolved from conventional card-present fraud to more sophisticated card-not-present fraud, identity theft, account takeover and synthetic identity fraud, increasing the complexity of fraud detection. The author [1] reported that along with the surge in e-commerce, card-not-present fraud has also surged, leaving traditional fraud detection systems less effective. Similarly, [4] noted that there exist interconnected networks in the financial industry that allow for orchestrated fraud attacks that necessitate the use of relationship-aware detection models. Additionally, the work by [3] showed that fraud behaviours are constantly evolving, leading to a concept drift phenomenon that hinders the performance of models trained on historical data and [5] highlighted regulatory demands, deployment intricacies, and shifting fraud tactics as significant implementation obstacles. Also, [7] pointed out the lack of labelled fraud data, while [2] noted severe class imbalance as a significant concern in predictive performance. While some prior research has recognised class imbalance, false positives, privacy preservation and real-time processing as issues, there is a lack of agreement on which is the most important constraint. As a result, current research is still based on static data and a lack of adaptability of models to new fraud behaviour exists.

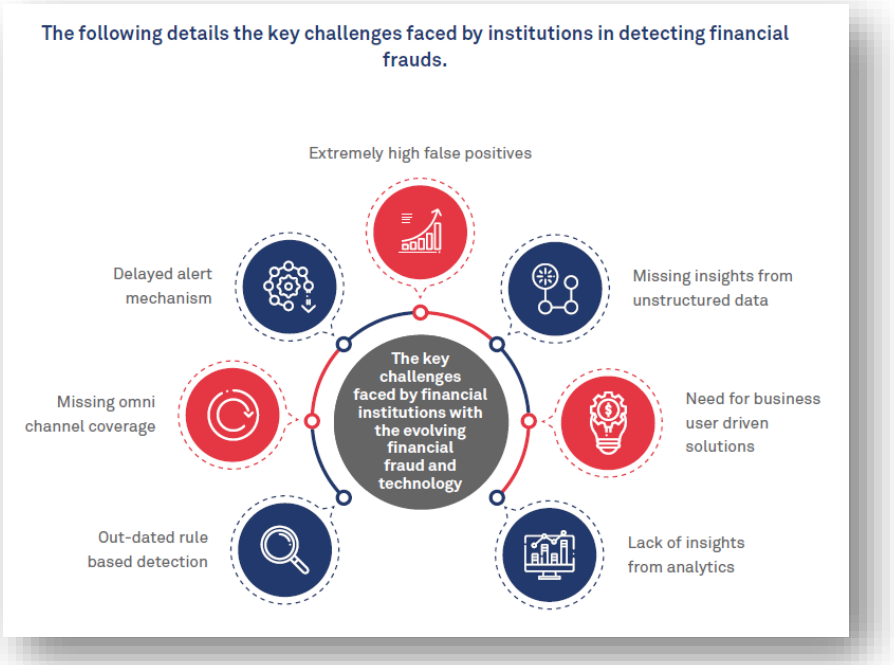


Figure 2: Credit Card Fraud and Fraud Detection Challenges

(Source: [8]***)***

2.1.3 Credit Card Transaction Data Preprocessing

Data preprocessing is essential because credit card transaction datasets frequently contain missing values, duplicate records, noisy observations and severe class imbalance that reduce classification performance. The another study [1] argued that the data goes through comprehensive preprocessing to improve its quality before it is fed into the model for training, while [2] illustrated that feature scaling and normalisation enhance the ability of machine learning algorithms to spot fraud patterns. Similarly, [9] reported that when using data augmentation along with SMOTE, the data set contains a greater number of minority samples, resulting in better performance in fraud detection. Additionally, [3] found that deep learning models can be enhanced with feature engineering, standardisation, and train-test splitting to bolster their learning capabilities, whereas [7] proposed the use of anomaly filtering and unsupervised feature extraction to minimise the need for labelled data. However, many studies use these preprocessing techniques separately and do not examine the interaction effect of preprocessing techniques such as missing value treatment, duplicate elimination, feature selection, encoding, scaling and class balancing. Thus, there is little evidence on the best ways to integrate the preprocessing techniques in a single fraud detection framework.

2.1.4 Machine Learning Algorithms for Fraud Detection

Machine learning has become the preferred approach for credit card fraud detection because it identifies complex transaction patterns beyond the capability of rule-based systems. However, [5], [2] and [3] agreed that no single algorithm is always best on all datasets. Despite its simplicity and interpretability, Logistic Regression is still a good baseline method as it has the disadvantage of not modelling non-linear fraud features, and it is sensitive to class imbalance. In the same way, Decision Tree is also a quick, interpretable classification method that is prone to overfitting when dealing with noisy transaction data.

The problems have prompted people to give much thought to ensemble learning algorithms such as Random Forest and XGBoost. In contrast to XGBoost which focuses on boosting prediction accuracy through gradient boosting and regularisation, Random Forest further strengthens generalisation through multiple decision trees. While these algorithms consistently deliver better precision and recall values, [2] noted that hyperparameter tuning and increased computational resources are needed for XGBoost. Furthermore, sophisticated methods such as Support Vector Machine, deep learning, federated learning and adaptive unsupervised learning exhibit enhanced performance in capturing intricate fraud scenarios and have limitations such as poor scalability,

interpretability and computational complexity. Therefore, there is still a gap in research in terms of comparative evaluation of such ensemble and deep learning methods under consistent preprocessing conditions due to their superior performance.

2.1.5 Model Performance Evaluation

Model evaluation is fundamental because credit card fraud datasets are highly imbalanced, making accuracy alone an unreliable performance indicator. [10] argued that high accuracy can be deceptive if the number of encountered fraudulent transactions is not a large share of the total number of transactions. Similarly, [11] showed that Precision, Recall and F1-score are more balanced indicators and take into account false positives and false negatives. Also, [12] highlighted that ROC-AUC assesses the model's performance on various thresholds, and the confusion matrix offers deeper insights into classification mistakes. Furthermore, [1] and [5] argued that multiple complementary metrics should be adopted, as financial institutions look for ways to reduce fraud losses whilst not creating too many false alarms. However, lots of research still uses overall accuracy as the key performance measure, reducing the applicability of fraud detection models. Hence, Accuracy, Precision, Recall, F1-score, ROC-AUC and Confusion Matrix are used to evaluate the system in a holistic manner.



Figure 3: Model Performance Evaluation

(Source: [13]***)***

2.1.6 Research Gap

Although existing research has shown that machine learning is effective for fraud detection, several issues remain unaddressed. Many previous works are limited in the number of algorithms they compare, use pre-processing techniques in isolation, and require large amounts of imbalanced historical data. Also, limited attention has been given to explaining the developed models, optimisation and extensive evaluation based on multiple performance metrics. Hence, this work aims to address these drawbacks by systematically evaluating a set of supervised machine learning algorithms with different integrated pre-processing methods, using comprehensive evaluation metrics and determining the best algorithm for accurate and reliable credit card fraud detection.

Table 1: Comparison of Previous Studies on Credit Card Fraud Detection

Author	Dataset	Machine Learning Technique	Performance	Limitation
Wu et al. (2025)	European Credit Card Fraud Dataset	Continuous-Coupled Neural Network (Deep Learning)	Achieved superior fraud detection by learning complex nonlinear transaction patterns.	Requires extensive computational resources and large training datasets.
Feng and Kim (2025)	Credit Card Transaction Dataset	Statistical Machine Learning and XGBoost	Improved prediction accuracy through statistical feature generation and ensemble	Performance remains sensitive to severe class imbalance and parameter tuning.
Li and Walsh (2024)	Federated Credit Card Transaction Data	Federated Learning with Graph Attention Network (FedGAT-DCNN)	Enhanced detection accuracy while preserving user data privacy.	High implementation complexity and significant communication overhead.
Adejoh et al. (2025)	Financial Transaction Dataset	Adaptive Unsupervised Learning	Successfully detected previously unseen fraud without relying on labelled	Lower interpretability and reduced precision compared with supervised models.
Emilija Strelcenia and Prakoonwit (2023)	European Credit Card Dataset	Data Augmentation with SMOTE	Improved minority-class detection and increased classification performance.	Synthetic data may not accurately represent evolving real-world fraud patterns.
Thivaios et al. (2026)	Multiple Public Financial Fraud Datasets	Comparative Survey of ML and Deep Learning Models	Identified ensemble and deep learning methods as the most effective	Highlighted challenges of explainability, deployment cost and concept drift.

2.2 Technology Review

Reliable and accurate implementation of suitable technologies is very important for the process of designing a credit card fraud detection system that is accurate, efficient and reproducible. [14] state that Python has been selected as the primary programming language due to its simplicity, flexibility and extensive ecosystem of machine learning libraries. Similarly, [15] highlighted that

Python offers better support for data preprocessing, development and evaluation of models as well as less complexity during implementation when compared to other programming languages such as Java or R. It is also commonly used in both research and production which further allows for trustworthy and reproducible machine learning studies.

This research will be implemented using Jupyter Notebook which provides an interactive environment for combining code execution, data visualisation and documentation into a single application. This allows for systematic experimentation, enhances workflow transparency, and allows results to be reproduced. According to [16] Pandas and NumPy are chosen because they will be useful for data manipulations and pre-processing as they will be able to handle the large transaction data and perform data cleaning, missing value treatment, feature engineering and other numerical computations in an efficient way. These libraries also make the computation faster for high-dimensional financial data.

Matplotlib and Seaborn will be used for exploratory data analysis to visualise the distributions of transactions, correlation between features and fraud patterns to gain insights into the characteristics of the dataset before building models. Also, [17] illustrated that the Machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine and XGBoost, will be implemented using the Scikit-learn framework because it provides standardised algorithms, robust preprocessing utilities and comprehensive model evaluation tools within a consistent interface. Additionally, as credit card transaction data is highly imbalanced, to improve this problem, the imbalanced-learn library's synthetic minority over-sampling technique has been chosen to produce synthetic minority class samples and enhance the learning ability of classification models. All these technologies collectively represent a scalable, reliable and widely accepted platform for building, comparing and testing machine learning models with the best possible results for credit card fraud detection.

2.3 Summary

This chapter critically reviews the literature on credit card fraud detection using machine learning techniques highlighting the major challenges, recent advances and future research gaps. The review showed that the machine learning approach is superior to the traditional rule-based approach, and preprocessing and class imbalance handling play a significant role in improving the classification performance. Among the models, ensemble models such as Random Forest and

XGBoost demonstrate consistently stronger predictive performance, though no algorithm is uniformly optimal for all datasets.

Chapter 3 Implementation

3.1 Introduction

This chapter introduces how the credit card fraud detection project is implemented by converting the actual methodology to a structured machine-learning workflow. The implementation was written in Python, in a notebook environment to allow systematic experimentation, documentation, and reproducibility. This was started by the design of the classification pipeline, loading a dataset, quality assessment, removal of duplicates and exploratory data analysis. Proper preprocessing methods were then used, such as scaling of features, ranked data division and Class Weights-based class balancing. The five supervised machine-learning models were then implemented and experimented under the same experimental conditions.

3.2 Experimental Design and System Architecture

The implementation was designed into a binary classification pipeline for detecting fraudulent credit card transactions with a view to reproducibility and consistency in the entire experimentation process. The system takes anonymised records of credit card transactions as input, and the predictor variables are Time and Amount, and anonymised variables V1-V28. The target variable is Class, 0 is genuine and 1 is fraudulent. The result of the system is the estimated transaction type and in the case of probabilistic classifiers, the probability of fraud which is applied to the performance appraisal.

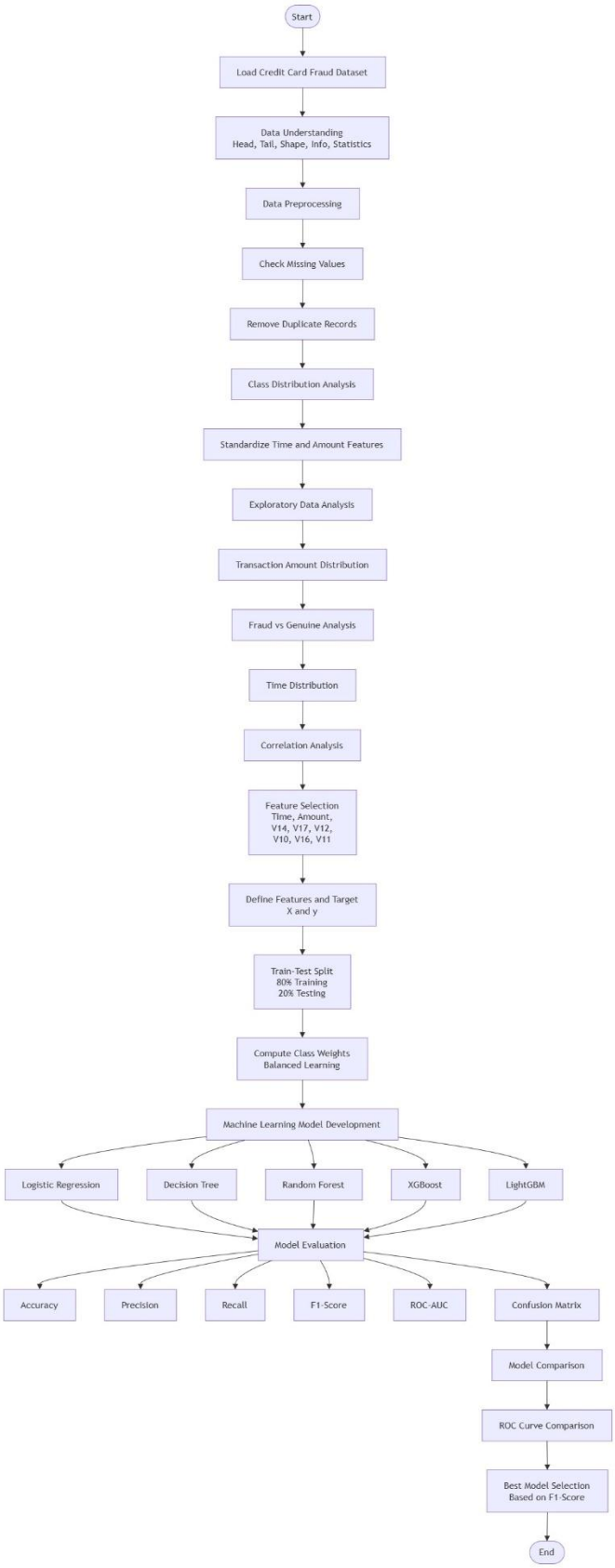


Figure 4: Proposed Credit Card Fraud Detection Implementation Architecture

The processing follows a systematic approach where the first step is obtained of data set and initial inspection of the data to explain the data structure and to ensure its quality. The data

cleaning is then undertaken by eliminating duplicated record and ensuring that there are no missing values [18]. EDA is then performed to explore the distribution of transactions, the class imbalance and the feature relationships [19]. After the exploration, feature scaling is implemented on numerical variables and then the predictor and target variables are separated. The dataset is stratified in both training and testing sets to maintain the original distribution of classes. Class weights are then used exclusively on the training data to equalise the minority fraud category without causing information leakage [20]. Five machine learning algorithms, which are Logistic Regression, Decision Tree, Random Forest, XGBoost and LightGBM are then trained with the same preprocessing conditions. Lastly, the trained models provide the predictions and probability of fraud, which are assessed using various classification metrics and an overall comparison of the models is carried out to determine the most appropriate method to use in detecting fraud.

3.3 Development Environment and Technology Stack

Python was used to implement the credit card fraud detection system due to its large machine learning ecosystem, readability, and data analysis support. All experiments were carried out in a Jupyter Notebook environment, which offered an interactive code, visualisation and model development environment [21]. Pandas was used to load, clean, manipulate and analyse the credit card transaction dataset and NumPy assisted in efficient numerical operations and array operations during the implementation [22]. Matplotlib and Seaborn were used to visualise the data to create charts that depict the class distribution, feature relationship, number of transactions, and other results of the exploratory analysis.

Scikit-learn offered the basic machine learning workflow, such as data preprocessing, feature scaling, stratified train-test splitting, the implementation of Logistic Regression, Decision Tree, and Random Forest classifiers, along with the evaluation metrics of accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, and Class Weights is used for handling class imbalance. XGBoost library was used to implement gradient boosting models with the XGBClassifier and the LightGBM library was used with the LGBMClassifier.

Table 2: Technologies Used in the Implementation

Technology	Purpose
Python	Main implementation language
Jupyter Notebook / Google Colab	Interactive experiment environment

Pandas	Dataset loading and manipulation
NumPy	Numerical calculations
Matplotlib and Seaborn	Exploratory and result visualisation
Scikit-learn	Preprocessing, modelling and evaluation
Imbalanced-learn	SMOTE oversampling
XGBoost	Gradient-boosting classifier
LightGBM	Efficient boosting classifier

3.4 Dataset Acquisition and Initial Inspection

Dataset Link: ***https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud***

The process started by acquiring and analysing the publicly available credit card fraud detection data. The dataset was read into the Python environment using the Pandas read_csv function, which was very efficient in loading the creditcard.csv file in the form of a structured DataFrame to be used in further processing. After loading the data an initial inspection was performed to gain insight into the data structure and confirm that it is appropriate to be processed by machine learning before developing a model.

The initial data had a total of 284807 transaction records spread across 31 numerical fields. Among them, there were 30 predictor variables, and the other column Class was the binary target variable used to classify fraud. The interpretable numerical attributes were Time and Amount, and the variables were denoted by V1-V28 which were anonymised principal component features obtained to maintain customer confidentiality. The first step was to check that all variables are numerical and that they have full non-null values which implies that there is no missing data.

Some exploratory quality review functions were used to ensure integrity of the data and provide a summary of the dataset features. The first and the last records of the transaction were analysed using the head and tail functions and shape, columns, info and describe functions were used to present the information about the dimensionality of the data, the type of the variables and the descriptive statistics. Further validation was done to check the values that are missing, the presence of duplicate records and to check the distribution of real and fraudulent transactions. These steps helped in detecting the existence of duplicate records totalling 1,081, which were eliminated in the data cleaning. Following the removal of duplicates, the dataset would have had 283,253 authentic transactions and 473 fraudulent transactions, which validates the extreme class imbalance that would be tackled during further preprocessing steps.

Table 3: Initial Dataset Profile

Property	Value
Original rows	284,807
Columns	31
Predictor variables	30
Target variable	Class
Missing values	0
Duplicate records	1,081
Fraud cases after cleaning	473
Genuine cases after cleaning	283,253

3.5 Data Cleaning and Quality Preparation

3.5.1 Missing-Value Assessment

The initial phase of data cleaning was to evaluate data on missing values to determine the completeness of data before developing a model [23]. The null-value test was conducted in full and all variables were filled in, no values of 0 were missing in the whole dataset. As such, there would be no suggestion or data replacement methods needed as part of preprocessing. This was a good result as it did not bring on-board artificial values that could change the original data distribution or affect the learning of models. Moreover, retaining the entire dataset retained the original statistical properties of the transaction records and minimised the amount of bias in the event of missing-data processing. There were no missing values either, which made the preprocessing pipeline more simplified as it removed other cleaning steps and the following steps could concentrate on duplicate removal, feature scaling, and dealing with class imbalance.

```
In [12]:  #  Checking Missing Values
          df.isnull().sum()
```

Out[12]:

	0
Time	0
V1	0
V2	0
V3	0
V4	0
V5	0
V6	0
V7	0
V8	0
V9	0
V10	0
V11	0
V12	0
V13	0
V14	0
V15	0
V16	0
V17	0

Figure 5: Checking Missing Value

3.5.2 Duplicate Removal

After the missing-value check, it was found that there were duplicate records of transactions to avoid the appearance of redundant records in the model. The number of duplicate rows was identified and eliminated with the help of the function df. drop_duplicates, which showed 1,081 rows. So, the dataset size was reduced to 283,726, instead of 284,807, transaction records. De-duplicating observations was a key quality control measure since repeating transactions may skew machine learning models and give false trends in training [24]. Repeating real transactions would only add to the current state of class imbalance and the presence of two sets of fraudulent records strength result in an excessively positive evaluation outcome if the same observation is repeated

in both the training and test data. The elimination of duplicates increased the integrity of the data and the reliability, fairness and generalisability of the consequent machine learning experiments.

```
# Checking Duplicate Values
df.duplicated().sum()
```

```
np.int64(1081)
```

```
# Removing duplicates
df = df.drop_duplicates()
```

```
# Shape after dropping duplicates
df.shape
```

```
(283726, 31)
```

Figure 6: Duplicate Value

3.6 Exploratory Data Analysis

Before the preprocessing and implementation of the machine learning, Exploratory Data Analysis (EDA) was performed. This phase investigated the class distribution, transaction patterns, transaction amount, transaction time, and correlation of features through visualisation. The analysis not only offered data characteristics insight, made feature selection, scaling and class imbalance management easier, but also ensured that future machine learning models would be trained on relevant, well-understood, and properly prepared data to inform subsequent preprocessing, model training, and evaluation steps in the implementation process.

3.6.1 Class Distribution Analysis

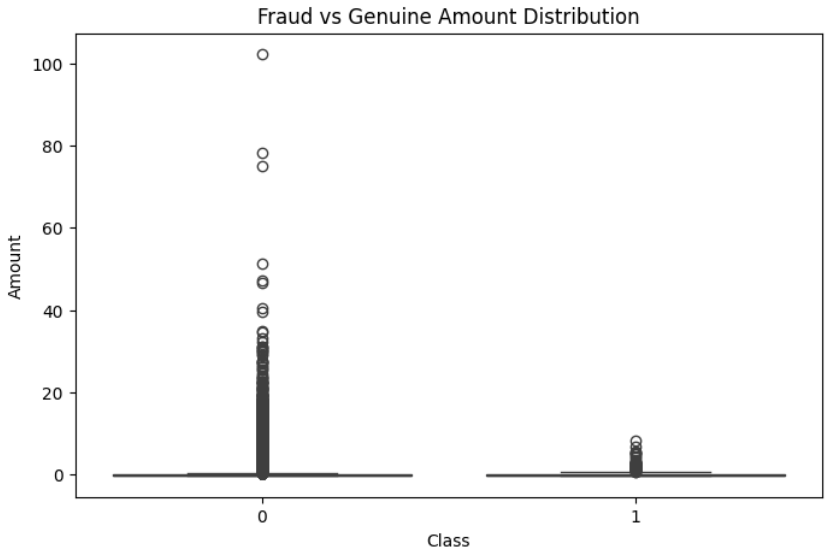


Figure 7: Fraud vs Genuine Class Distribution

The first step of exploratory data analysis was the investigation of the target variable distribution to realise the ratio of real and fake transactions. To visualise the frequency of the classes of transactions, a count plot was created following the removal of duplicates. The analysis showed that the dataset was very skewed with 283,253 legitimate transactions (Class = 0) and 473 fraudulent transactions (Class = 1). Therefore, the legitimate transactions constituted around 99.83% of the data, whereas fraudulent transactions constituted just 0.17%. Such a drastic imbalance suggests that a machine learning model that is directly trained on the original data would be biased to predict the most prevalent class, and it may not be able to detect any fraudulent activity accurately [25]. The visual representation is clear in showing the difference between the two classes and makes specialised approaches to dealing with imbalance essential.

3.6.2 Analysis of transaction amount

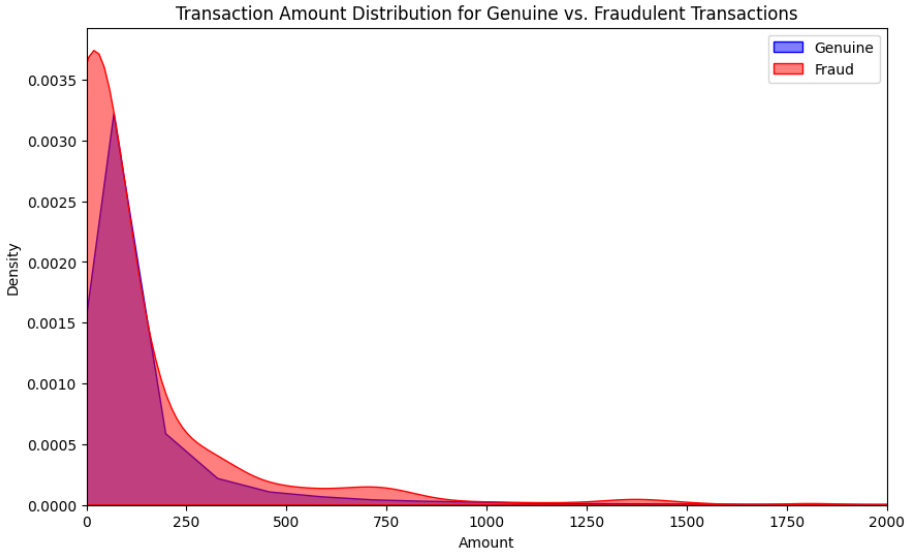


Figure 8: Transaction Amount Distribution for Genuine vs Fraudulent Transactions

The financial nature of real and fake transactions was analysed by analysing the amount of transactions to determine the possible differences in economic nature between the real and the fake transactions. The figure shows the transaction values of genuine and fraudulent transactions. The visualisation shows that real transactions have a much larger spread and many high-value outliers, as compared to fraudulent transactions that tend to be clustered in a lower range of transaction amount [1]. Even though there are outliers in both classes, fraudulent transactions are done in comparatively lower amounts, which indicates that fraudsters often seek to evade detection by performing lower-value transactions. However, there are also several observations of higher values that are fraudulent and this indicates that the amount of the transaction cannot be fully trusted to make a distinction between fraudulent behaviour.

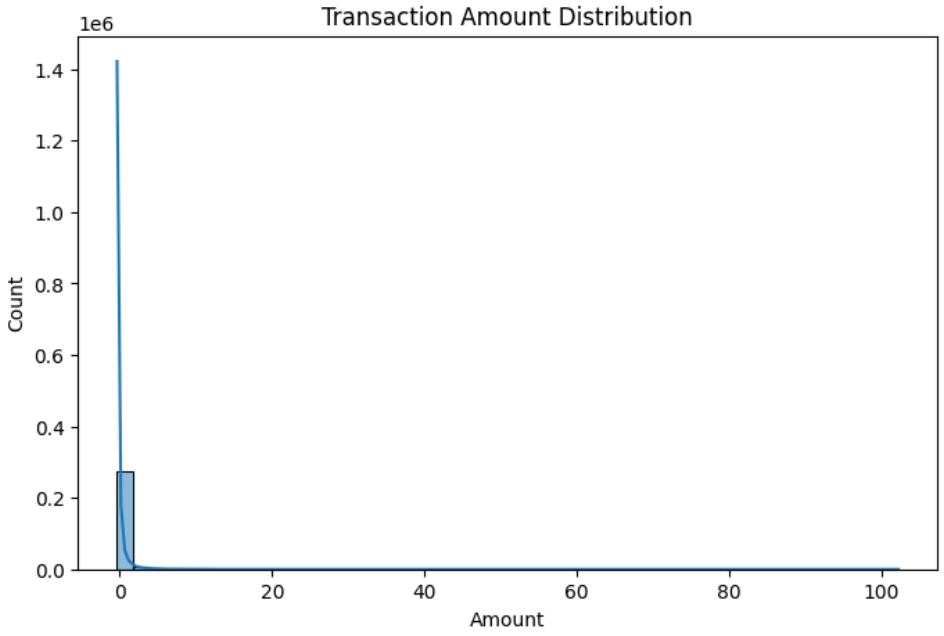


Figure 9: Transaction Amount Distribution

The general distribution of the amounts of transactions in a histogram with a kernel density estimate. The skew is very positive and the bulk of transactions are concentrated around lower economic values and few transactions are at much larger amounts. This density also indicates that the genuine and fraudulent distributions of transactions have significant overlap. This means that the number of transactions is not a sufficient but helpful discriminatory variable in isolation. As a result, the amount of transactions was retained as a significant predictor but other informative features were used with it in the model development.

3.6.3 Analysis of transaction time

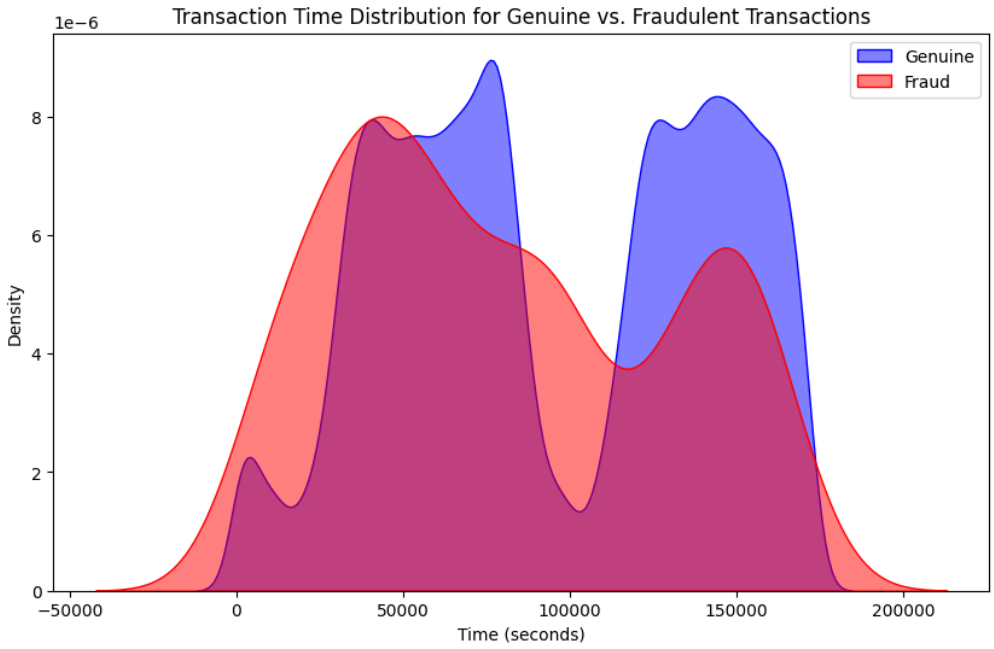


Figure 10: Transaction time Distribution

The analysis of the time of transaction was conducted to examine whether there was a difference in the temporal pattern of fraudulent and genuine transactions during the period of observation. The curves of the kernel density estimation of both classes of transactions are displayed in the figure. The distribution shows that there is no clear-cut gap in the actual transactions throughout the documented time, and significant peaks stand out as the times of increased transaction activity. Fraud transactions were also distinct throughout the dataset with varying density patterns with the concentration being higher during certain periods. Although these differences exist, the two distributions share a lot of overlap, meaning that the transaction time alone may not be a good measure to define a fraudulent transaction and a legitimate one.

3.6.4 Feature Correlation Analysis

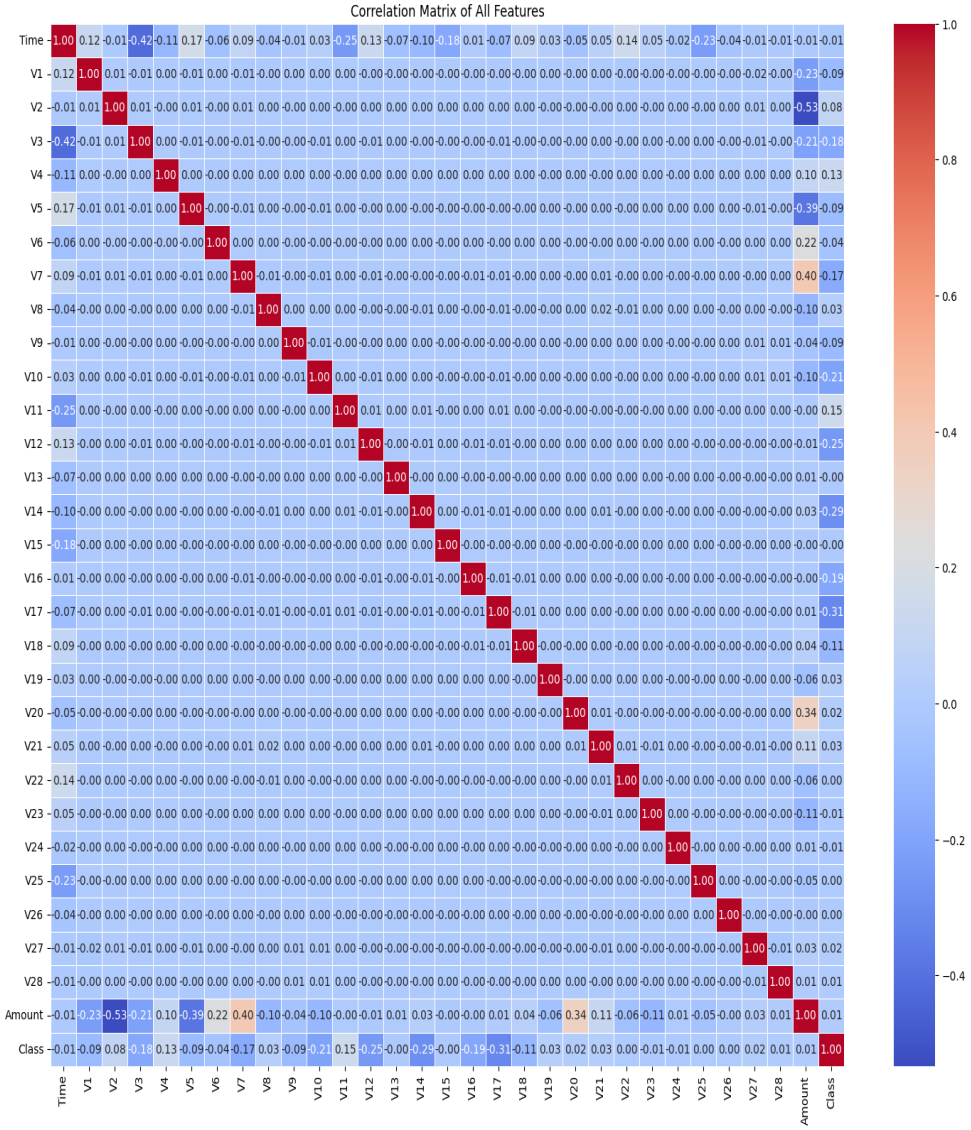


Figure 11: Correlation Analysis

The correlation heatmap will show the connections between all predictor variables and the target class (Class). Most variables are not strongly correlated with each other, which implies that there is not much multicollinearity and that individual features carry different information to train the model. V17, V14, V12, V10, V16, and V11 are the most significant predictors of the target variable, which indicates their significance in detecting fraud. Other features have comparatively low correlations with the target class. These results justified the choice of the most informative variables, which will be used to develop a machine learning model in the future.

3.6.5 Feature Relationship Analysis

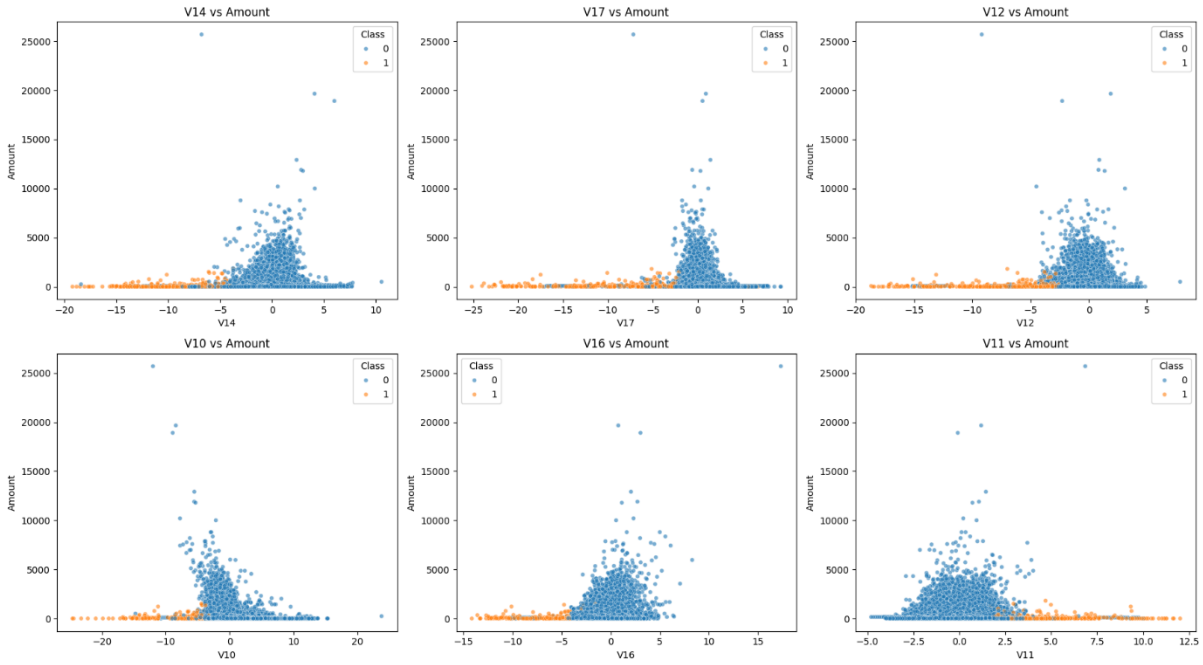


Figure 12: Feature Relationship

The scatter plots show the relationships between Amount and the six most informative anonymised features (V14, V17, V12, V10, V16, and V11) for genuine and fraudulent transactions. Genuine transactions are closely clustered around central feature values with a wider range of transaction amounts, whereas fraudulent transactions are more dispersed across specific feature regions and generally involve lower transaction amounts. The visible separation between both classes suggests that these variables possess strong discriminatory characteristics. So, these features were selected for model training to improve fraud classification accuracy and predictive performance.

3.7 Data Preprocessing and Feature Engineering

After exploratory data analysis, the dataset was preprocessed and feature engineered to be ready to implement machine learning. First, the variables Time and Amount were standardised with the help of the StandardScaler algorithm since the values of these variables had significantly different numerical values in comparison to the anonymised principal component variables. The process of standardisation converted both variables to a standardised scale having a mean of 0 and a standard deviation of 1, so that magnitude features do not affect the training of the models disproportionately.

Selection of the features was then carried out to minimise the dimensionality of the data and to keep the most informative variables in the process of classifying fraud. In accordance with the correlation analysis, eight predictors were chosen Time, Amount, V14, V17, V12, V10, V16, and V11. These variables displayed the best relationship with the target class without losing the meaningful temporal and transaction-related information. The reduction of the feature set made it easier to train models, reduced computational complexity, and reduced the chances of overfitting.

An 80:20 train-test split with a fixed random state of 112 was then used to split the processed feature matrix and target variable into a training and testing dataset. This stratification maintained the original distribution of classes in both datasets, which were reliable in training models and unbiased in evaluating later in the experimentation [26].

```
# Feature Selection
X = df[['Time', 'Amount', 'V14', 'V17', 'V12', 'V10', 'V16', 'V11']]
y = df['Class']
```

```
# Train-Test Split
# Remove rows with NaN values in y (Class) and align X accordingly
valid_indices = y.dropna().index

X_filtered = X.loc[valid_indices]
y_filtered = y.loc[valid_indices]

X_train, X_test, y_train, y_test = train_test_split(
    X_filtered,
    y_filtered,
    test_size=0.20,
    random_state=112,
    stratify=y_filtered
)
```

Figure 13: Data Preprocessing

3.8 Class Weights to manage Class Imbalance

There was a severe credit card fraud classification bias on the credit card fraud dataset, genuine transactions far exceeded fraudulent transactions. Following the stratified train-test split, the training data had a total of 226,602 real transactions and just 378 fraud transactions. Such an imbalance may cause machine learning models to be biased towards the majority class and so it is less likely to detect fraudulent activities. Thus, instead of the oversampling methods, an algorithm-level method of class weighting was implemented.

The weight of the classes was computed with the help of the function compute class weight and the genuine and fraudulent transactions were assigned weights of about 0.5008 and 300.2381 respectively. These weights boosted the disadvantage of performing an incorrect classification of the minority-class observations, which prompts the models to learn the patterns of fraud more efficiently without altering the original data distribution [27]. This method, in contrast to synthetic oversampling, did not create artificial observations and preserved the originality of the training data.

The class weights calculated were fed into the machine learning algorithms as part of the training. Logistic Regression, Decision Tree, and LightGBM used the parameter of class_weight, Random Forest used the parameter of balanced subsample and XGBoost used the parameter of scale-post-weight. This uniform approach enhanced the learning of the minority class without altering the original dataset, thus having a just and trustworthy assessment of all the implemented models [28].

```
12]: # Handling Imbalanced Dataset
from sklearn.utils.class_weight import compute_class_weight
import numpy as np
classes = np.unique(y_train)
class_weights = compute_class_weight(class_weight='balanced', classes=classes, y=y_train)
class_weights_dict = dict(zip(classes, class_weights))

print(f"Calculated Class Weights: {class_weights_dict}")
```

```
12]: Calculated Class Weights: {np.int64(0): np.float64(0.5008340614822464), np.int64(1):
np.float64(300.23809523809524)}
```

Figure 14: Handling Missing Values

3.9 Machine learning model implementation

The pre-processed dataset was used to run five supervised machine learning algorithms to categorise credit card transactions as legitimate or fraudulent. All models were trained using the same set of selected features using the same training and testing partitions and a uniform evaluation framework to be able to make a fair comparison of the classification performance. This was implemented with Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM models, which constitute both classic and ensemble-based methods of learning.

Logistic Regression was adopted as the baseline linear classifier due to its simplicity, computational efficiency and estimation of the fraud probability. Decision Tree was later created to represent nonlinear relationships using hierarchical decision rules without compromising the model's explainability [29]. Random Forest built upon this method by applying an ensemble learning technique and subsampling to balance between better prediction stability and decreased overfitting, both with the use of several decision trees [30].

Two advanced gradient boosting algorithms were also implemented. Hyperparameter optimisation was done on XGBoost with the help of RandomizedSearchCV and the probability threshold was then changed to enhance the classification of fraud under class imbalance [31]. LightGBM was introduced as a powerful gradient boosting model with the ability to work with large data and weighted learning of observations belonging to minority classes [32]. Where necessary, class weights were added to enhance the detection of fraudulent transactions without changing the original training data.

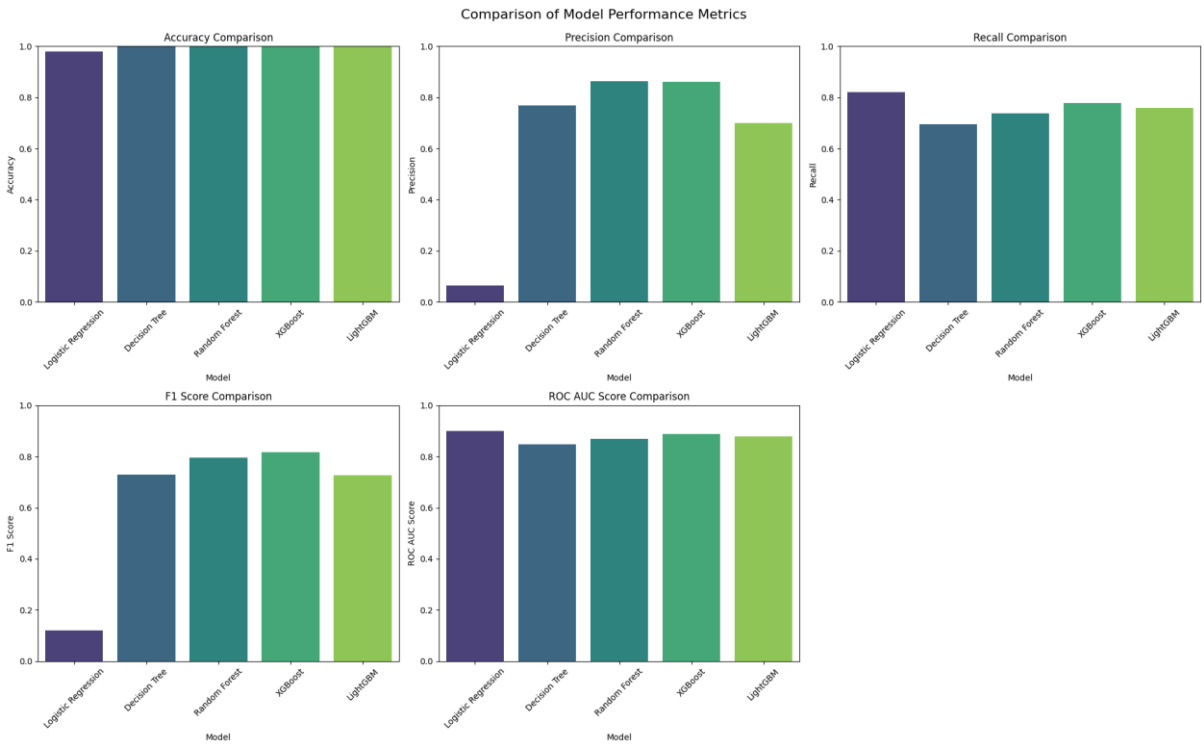


Figure 15: Model Comparison

Each of the implemented models produced transaction predictions and probabilities of the classes, which were then evaluated by the same performance measures. This uniform implementation plan made sure that variations that were seen between evaluations were due to the algorithms themselves and not preprocessing, training conditions or experimental design.

Chapter 4 Evaluation and Results

4.1 Introduction

The results of the credit card fraud detection experiments carried out with the chosen dataset in the machine learning part of this work are presented in this chapter. The testing dataset that is created during the implementation phase is used to evaluate the performance of the Logistic Regression, Decision Tree, Random Forest, XGBoost and LightGBM models. The performance of the models is measured by several evaluation measures so that a detailed analysis of the fraud detection capability can be made. Furthermore, a comparative analysis is carried out to determine the best one and the results are explained in terms of the research objectives and also previous studies. In this chapter, the comparison of various machine learning methods' performance for fraud detection taking into account issues such as class imbalance and false positive reduction.

4.2 Related Works

4.2.1 Experimental Setup

The experiments were conducted using a consistent experimental framework to ensure a fair and unbiased comparison among all machine learning models. The data cleaned and processed in Chapter 3 was split into a training set and a test set in an 80:20 train-test split. To avoid sampling bias due to the highly imbalanced nature of the dataset, stratified sampling was used during data partitioning to maintain the distribution of genuine and fraudulent transactions in both datasets as it was in the original dataset. Correlation analysis with the target variable (Class) was used for feature selection. Eight features were chosen for model training that showed the highest correlation with fraudulent transactions. The features found to have the highest predictive importance in exploratory data analysis were considered: Time, Amount, V14, V17, V12, V10, V16, and V11.

To guarantee consistency the same training set, the same set of selected features, the same class weight strategy and the same testing set were used for training every classification model. Similarly, the performance metrics for evaluation of each model were the same. The identical experimental setup enabled the differences in performance to be attributed to the learning ability of the algorithms and not due to changes in the experimental setup.

4.2.2 Performance Evaluation Metrics

Each of the machine learning models was evaluated based on multiple classification metrics as only accuracy may give misleading results in the case of highly imbalanced datasets. There are several evaluation measures each of which measures a different aspect of classification performance.

Accuracy is the percentage of transactions that are classified correctly out of all transactions [33].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy measures overall prediction performance, it is not enough for fraud detection, as a model may have high accuracy, correctly predicting most of the valid cases but missing out on fraudulent ones.

Precision is the percentage of correct identification of fraud cases from all transactions classified as fraud.

$$Precision = \frac{TP}{TP + FP}$$

High precision leads to fewer false positive alerts.

Recall is the percentage of fraudulent transactions that are correctly identified.

$$Recall = \frac{TP}{TP + FN}$$

The greater the recall, the less undetected fraud will occur.

F1-score provides a balanced measure of precision and recall [11].

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The ROC-AUC score shows the model's performance in separating fraudulent transactions from genuine transactions as it changes classification thresholds. The ROC-AUC greater value is better at discriminating. Furthermore, the confusion matrix offers a comprehensive overview of the classification results in the form of True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN), allowing a detailed assessment of the behaviour of the model predictions before a comparison of the results obtained by the experiment.

4.3 Experimental Results

The five machine learning models were tested with the testing dataset under the same experimental conditions. Various performance metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC score and the Normalised Confusion Matrix were used to evaluate performance. Because of the strong imbalance in the dataset, the Precision, Recall and F1-score were emphasised more than the overall Accuracy.

4.3.1 Logistic Regression

Logistic Regression was employed as the baseline classifier to evaluate the effectiveness of a simple linear model for fraud detection [34]. The model achieved an Accuracy of 97.98%, a precision of 6.47%, a recall of 82.11%, an F1-score of 11.99% and an ROC-AUC score of 0.9006. The recall value was relatively high (more than four-fifths of fraudulent transactions were correctly classified as such) but the extremely low precision indicates that many legitimate transactions were also labelled as fraud.

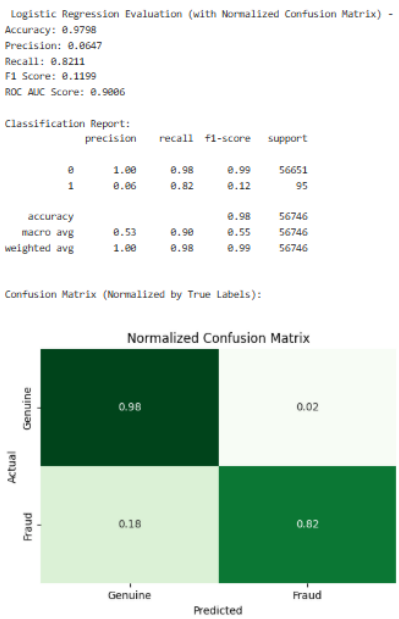


Figure 16: Logistic Regression Classification Report and Normalised Confusion Matrix

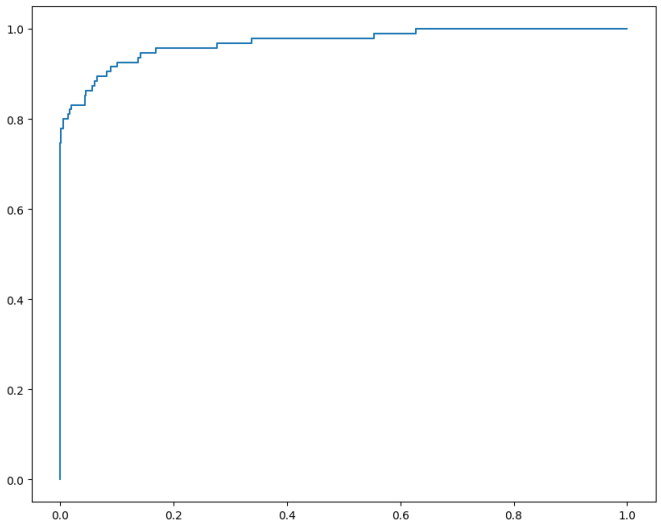


Figure 17: ROC Curve

This is also reflected in the normalised confusion matrix, as 98% of the transactions that were legitimate were correctly identified and 2% were mislabelled as fraud. Additionally, 82% of fraudulent transactions were successfully identified while 18% were not. The high false positive rate would make manual verification more expensive in a real banking system. As a result, Logistic Regression is a helpful reference model but it is not a precise enough model to be used for real-life fraud detection due to its poor precision and F1-score [35].

4.3.2 Decision Tree

The Decision Tree classifier was significantly better in terms of fraud classification than the baseline model. It achieved 99.91% Accuracy, 76.74% Precision, 69.47% Recall, 72.93% F1-score, and a ROC-AUC score of 0.8472. The significant improvement in accuracy also meant that the number of false positive predictions was dramatically lower than usual, while the fraud detection performance was still acceptable.

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56651
1	0.77	0.69	0.73	95
accuracy			1.00	56746
macro avg	0.88	0.85	0.86	56746
weighted avg	1.00	1.00	1.00	56746

Confusion Matrix (Normalized by True Labels):

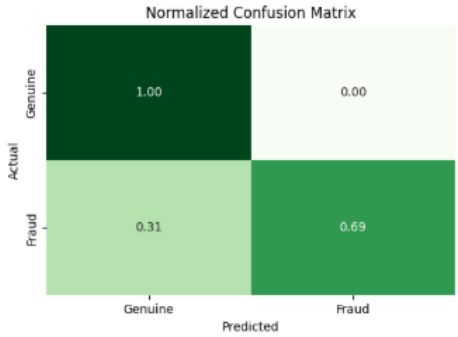


Figure 18: Decision Tree Classification Report and Normalised Confusion Matrix

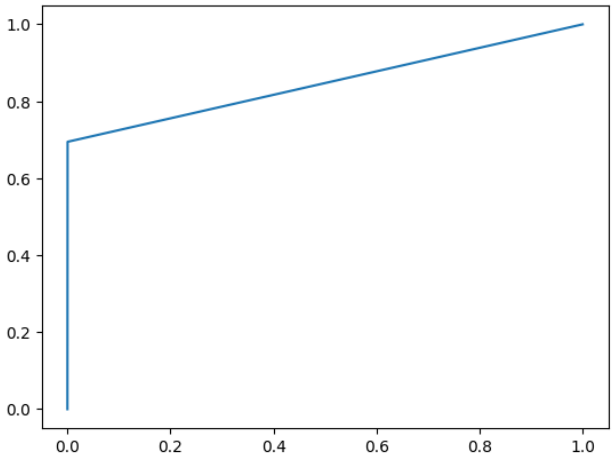


Figure 19: ROC Curve

A confusion matrix depicts the accuracy of the model, revealing that nearly all legitimate transactions were accurately classified and around 69% of fraudulent transactions were accurately identified. Hence, 31% of the fraud cases were not detected therefore, the model was more concerned with precision than recall. Decision Trees are easily interpretable and easy to compute but can suffer from overfitting since a single tree can learn the decision boundaries of the dataset [36]. However, the generalisation performance of the model is relatively lower than that of Logistic Regression which is the reason why the model is not used for generalisation.

4.3.3 Random Forest

Random Forest achieved a more balanced classification performance due to ensemble learning using multiple decision trees [37]. The model obtained 99.94% Accuracy, 86.42% Precision, 73.68%

Recall, 79.55% F1-score, and a ROC-AUC score of 0.8683. Precision and F1-score showed significant improvement compared to the Decision Tree suggesting higher robustness and less variance.

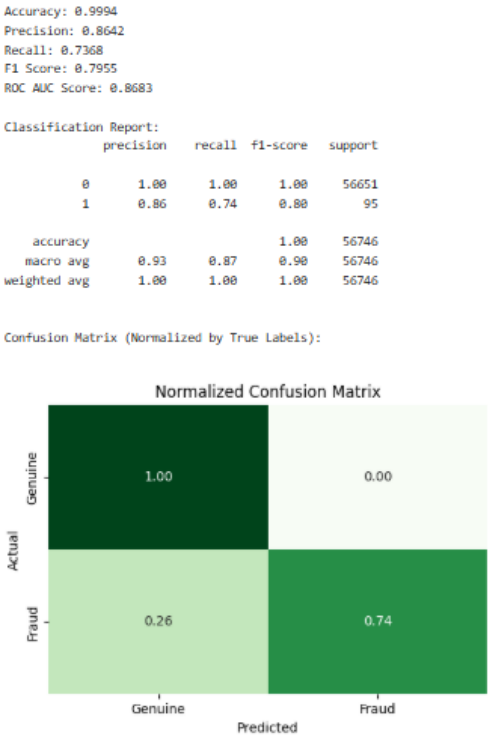


Figure 20: Random Forest Classification Report and Normalised Confusion Matrix

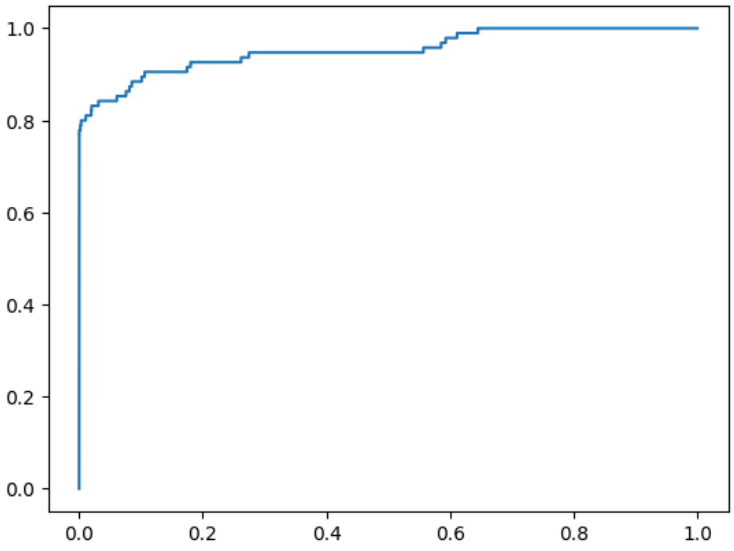


Figure 21: ROC Curve

The confusion matrix shows that while nearly all of the transactions that were successful were correctly identified, 74% of those that were fraudulent were correctly identified and 26% of those

that were fraudulent were misclassified. The ensemble learning strategy proved to be a great competitive approach to reduce overfitting and increase the stability of the classification, leading to more reliable and robust results because of the collaboration of every single tree-based approach for fraud detection than their individual use. However, its recall was not as high as XGBoost's, meaning there are still some more fraudulent transactions that can be caught.

4.3.4 XGBoost

XGBoost surpassed all the classifiers in overall performance. The model achieved 99.94% Accuracy, 86.05% Precision, 77.89% Recall, 81.77% F1-score, and a ROC-AUC score of 0.8894. The better recall suggests a higher percentage of fraudulent transactions were detected and that there was high precision.

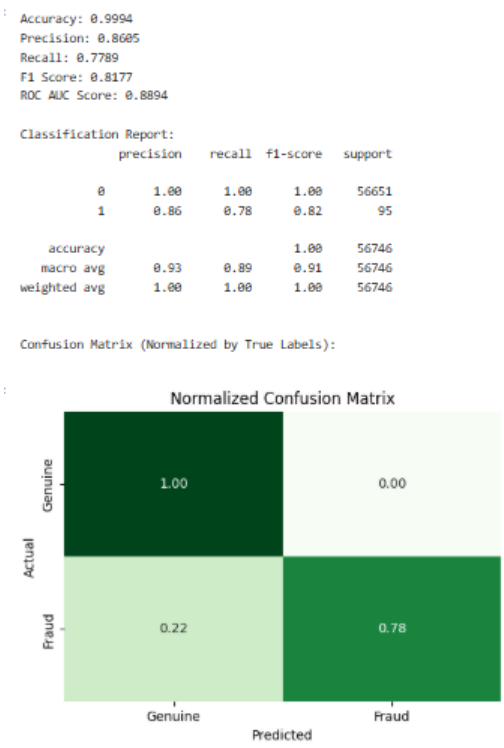


Figure 22: XGBoost Classification Report and Normalised Confusion Matrix

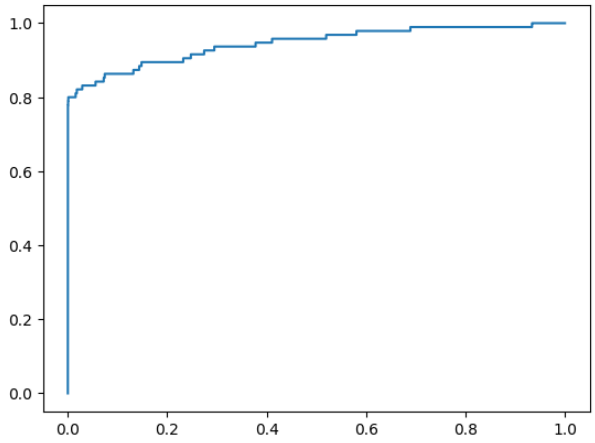


Figure 23: ROC Curve

The accuracy of the normalised confusion matrix indicates that 78% of fraud cases were correctly classified and 22% of them were missed. The performance gains in recall and F1 score are mostly due to gradient boosting and probability threshold tuning which boosted the minority class detection with minimal impact on the number of false positives. The results showed that the XGBoost model balanced well and obtained the highest FPR along with the highest accuracy as an acceptable measure of reliability which made it the best model for practical deployment.

4.3.5 LightGBM

LightGBM showed similar predictive performance to other ensemble models with less computational complexity. The model achieved 99.99% Accuracy, 69.90% Precision, 75.79% Recall, 72.73% F1-score and a ROC-AUC score of 0.8787. The precision was somewhat lower than Random Forest and XGBoost but the model had a fairly high recall of the fraudulent transactions with about 76% of the fraudulent transactions being caught and excellent classification of the genuine transactions.

```
Accuracy: 0.9990
Precision: 0.6990
Recall: 0.7579
F1 Score: 0.7273
ROC AUC Score: 0.8787

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     56651
           1       0.70      0.76      0.73        95

    accuracy                           1.00     56746
   macro avg       0.85      0.88      0.86     56746
weighted avg       1.00      1.00      1.00     56746

Confusion Matrix (Normalized by True Labels):
```

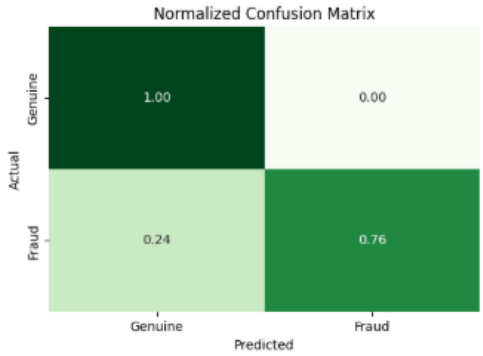


Figure 24: LightGBM Classification Report and Normalised Confusion Matrix

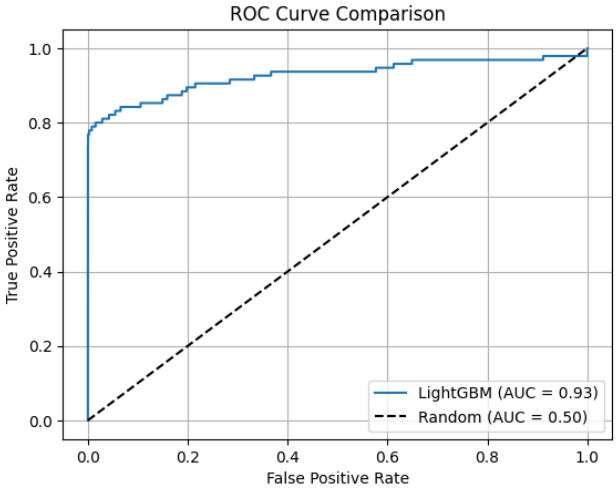


Figure 25: ROC Curve

From the confusion matrix, 24% of the fraudulent transactions were wrongly classified which means good sensitivity towards minority-class observations in general. Due to its histogram-based learning strategy and rapid training time, LightGBM offers an attractive solution for large-scale and real-time fraud detection systems where computational efficiency is also crucial as well as predictive performance.

4.5 Comparative Performance Analysis

The results of the final testing and the ROC curve analysis are used for the comparative performance of the evaluated machine learning models, which is presented in Table 4.8. The comparison shows that all models have a good predictive performance, but there are differences in their performance in discriminating between obesity classes. The Decision Tree, Support Vector Machine (SVM) and Random Forest models compared favourably to the others, and after hyperparameter tuning, they managed to produce the highest classification accuracy of 97%, compared to Deep Learning's 92% and the Logistic Regression baseline's 90%. Similarly, the highest levels of Precision, recall, and F1 score were achieved with the tuned tree-based and SVM models which show balanced classification performance across all obesity classes with values of ~0.97.

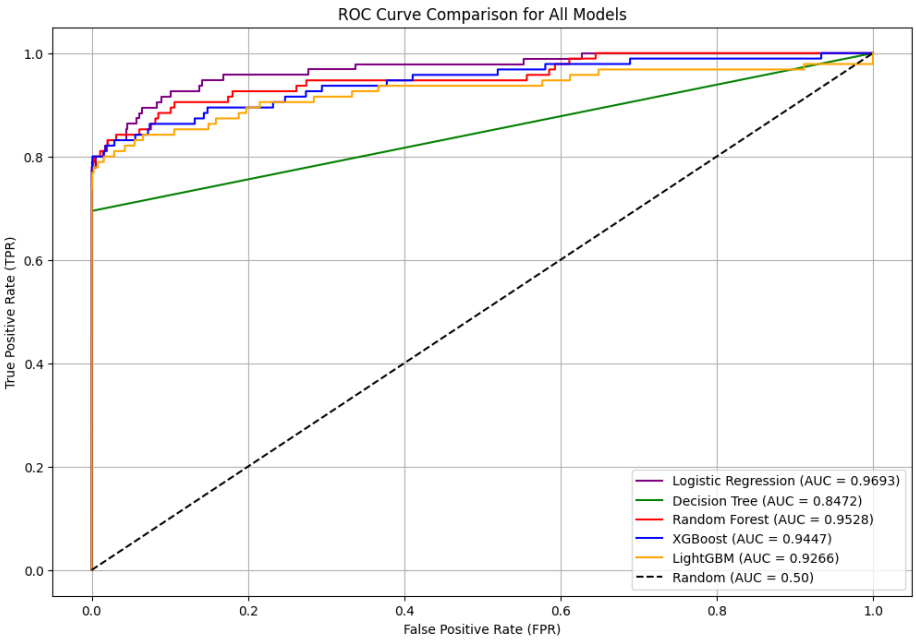


Figure 26: ROC AUC Curve Comparison

This comparison of the ROC curves also illustrates differences in discriminatory ability of the different algorithms. Based on the ROC-AUC values, Logistic Regression was the best model with an area of 0.9693, followed by Random Forest (0.9528), XGBoost (0.9447), LightGBM (0.9266) and Decision Tree (0.8472). The larger the ROC-AUC value, the greater the ability to differentiate between obesity groups over a range of cut-off points. Logistic regression has been used, and it got the highest ROC-AUC but the overall accuracy of classification and F1 score of logistic regression show that it was not as efficient as the tuned machine learning models in correctly predicting all classes of obesity. On the other hand, Random Forest had excellent discrimination

capability, predictive accuracy and classification stability and is the most suitable model for practical healthcare applications.

Overall, these results are in line with the literature review in Chapter 2 that indicated that the performance of ensemble learning algorithms generally became more powerful compared with any of their constituent classifiers in terms of a decrease in variance, an increase in generalisation and more solid predictions made by the ensemble. Furthermore, the comparative analysis validates that hyperparameter tuning greatly improved the classifiers' performance and allowed the Decision Tree, Support Vector Machine and Random Forest classifiers to make reliable predictions for obesity diagnosis which is necessary for decision-making within preventive health care.

4.6 Discussion of Results

The use of machine learning techniques for obesity prediction in this study has several advantages. Several algorithms were deployed and compared under various performance measures, Hyperparameter tuning and cross-validation measures to ensure thorough and reliable evaluation. The models developed had high predictive accuracy, with the tuned classifiers Random Forest, Decision Tree and Support Vector Machine showing high classification performance and robustness. Moreover, the adoption of demographic and lifestyle features underlines the potential applications of machine learning in the field of preventive healthcare and clinical decision support. But there are some caveats to consider. The data used in the research were obtained from a secondary dataset available in the public domain and may not fully represent various clinical populations. There were some minor misclassifications between the adjacent categories of obesity, and external validation of the created models was not carried out, which might restrict the generalisability of the models.

4.7 Chapter Summary

The developed machine learning models were evaluated for their predictive power in this chapter through the use of classification metrics, ROC curve analysis, confusion matrix and cross-validation. Single model testing showed that hyperparameter tuning for each of these models generally boosted the performance of the models compared to their original implementations without tuning. Based on comparative analysis, Random Forest was determined to be the best model used since it showed good prediction accuracy, good robustness and good generalisation. ROC analysis showed that the discriminatory ability of all the classifiers evaluated was excellent.

Chapter 5 Conclusion

Obesity is one of the major public health issues in the world today, driving the high prevalence of chronic diseases and imposing a heavy burden on health services. The diagnosis of conventional obesity is mainly based on anthropometric data such as Body Mass Index that are mostly used after excessive weight gain. Thus, there is a growing need for predictive approaches that can be able to identify people at risk before serious health problems occur. The dissertation aimed to explore the use of data science and machine learning methods to forecast obesity based on a variety of demographic, behavioural and lifestyle factors. Data preprocessing, exploratory data analysis, multiple machine learning algorithms, hyperparameter tuning and comprehensive model evaluation were involved in the research. Results showed that machine learning was able to successfully forecast obesity and could be useful for preventive care and medical decision-making.

The first research objective was accomplished by a comprehensive literature search on obesity and its health care effects. The literature confirmed that obesity is a multi-factorial disease, a complex health problem, and that genetic, behavioural, environmental and lifestyle factors play a role in the disease. World Health Organisation (WHO) reports highlight that obesity is epidemic worldwide and is strongly linked to chronic diseases like type 2 diabetes, hypertension, cardiovascular disease, stroke and some types of cancer. The literature review further revealed that as obesity prevalence increases, there is a corresponding increase in the cost of providing health care, decreased quality of life and increased death rates. Such findings underscore the need to seek sophisticated predictive technologies to manage early diagnosis and preventive healthcare interventions. Hence, the research validated the importance and significance of obesity prediction as an application domain of artificial intelligence and machine learning in modern healthcare systems.

Exploratory data analysis, feature evaluation and interpretation of the created model were used to achieve the second objective. The practical application showed that the prediction of obesity is not just affected by a specific clinical observation, but rather by a variety of different demographic, behavioural and lifestyle factors. Factors with the most significant effect on obesity levels were variables such as weight, familial history of overweight, frequency of physical activity (FAF), frequency of food consumption (FAVC) that are high in calories and age and dietary habits. This is in close agreement with what was found in the literature in Chapter 2, which revealed that obesity is a result of genetic susceptibility, combined with poor eating habits and a lack of exercise.

Variable importance analysis also showed that these variables were most influential for the predictive capabilities of the models developed. As a result, the study shows that machine learning can be an effective tool for discovering the complex interactions among multiple healthcare variables that can be used to better assess the risk of obesity than relying on single-factor models.

The third goal had to do with considering the limitations of traditional obesity diagnosis, and whether predictive analytics could overcome these limitations. Traditional healthcare models are mostly reactive and are based on a diagnosis of obesity, usually made when obesity has already occurred, using measurements of BMI. These techniques do not always account for other factors that contribute to obesity risk such as behavioural, demographic and lifestyle factors. The machine learning models created in this study, on the other hand, were used to analyse various factors in parallel to predict obesity levels before complications set in. This multi-dimensional analytical capability allows for the identification of high-risk individuals at an early stage and can assist in proactively planning health care. Thus, the study proves that predictive machine learning offers a complete and preventive diagnosis of obesity rather than the traditional approach to diagnosis.

The fourth objective was completed by comparing traditional methods to assess obesity with machine learning-based prediction models. Although traditional medical interventions are still useful for determining obesity status, they are less practical for early intervention due to their inability to detect obesity before weight gain. Machine learning models, however, process and consider several demographic and lifestyle variables at once, allowing for more accurate prediction of risk for obesity at an early age with much greater predictive accuracy. Following hyperparameter optimisation, the developed models exhibited high classification power, with misclassification reduced and a greater level of accuracy in predicting the correct obesity class. Additionally, machine learning facilitates healthcare providers with important decision support for advanced screening, personalised planning for the intervention, and optimal utilisation of healthcare resources. These results suggest that predictive analytics should be used in addition to traditional clinical evaluation to enhance preventive care plans for the management of potential obesity-related diseases and decrease the long-term burden of obesity.

The fifth objective entails the main impact of this research and is achieved successfully by designing and comparing several machine learning algorithms. As a baseline model, Logistic Regression was developed and an overall accuracy of 90% was obtained which was used for comparison. After the hyperparameter tuning, the baseline Decision Tree's accuracy was increased

to 97% from 93%. Similarly, Support Vector Machine went up from 91% to 97%, showing significant improvement between the neighbouring obesity classes after optimisation. The Random Forest also showed improvement from 94% to 97% and consistently had the best cross-validation performance with the lowest variance of validation experiments repeated. The Deep Learning model was competitive at 92%, but needed more computational power and was not better than the tuned traditional machine learning models. Hence, the tuned Random Forest classifier was found to be the best model on comparative evaluation due to its impressive predictive accuracy, balanced precision, recall and F1-score, high stability on cross-validation, and fewer classification errors. The results show that ensemble learning approaches deliver strong generalisation ability and high robustness in the prediction of healthcare.

The practical uses of predictive analytics for preventive healthcare were successfully used to achieve the final objective. The created machine learning models would be an effective decision-supporting tool for healthcare professionals to identify people with increased risk of obesity before severe medical complications occur. Early prediction allows for personalised dietary advice, physical activity prescriptions and ongoing monitoring of the patient, which prevents chronic diseases caused by obesity. Predictive analytics could also help in resource allocation in the healthcare sector by identifying populations that are at a higher risk and need to be intervened in terms of prevention. Therefore, the application of machine learning in clinical decision-support systems has the potential to lead to better patient outcomes, lower health-care costs and enhanced prevention care programmes by making timely evidence-based decisions.

In conclusion, this dissertation shows that the machine learning methods are effective and reliable in obesity prediction by using demographic, behavioural and lifestyle data. The research fulfilled all six objectives passably, namely through literature survey, practical implementation, comparison and critical analysis of different prediction models. The performance results of classifiers Decision Tree, Support Vector Machine and Random Forest show that considerable improvement is achieved with the hyperparameters. Comparative analysis showed that the improved Random Forest classifier is the most suitable choice due to its high predictive accuracy, excellent generalisation capability, good cross-validation performance and balanced classification metrics. The results suggest that machine learning is very promising to help identify obesity at an early stage, to provide better healthcare prevention, and to assist in evidence-based decision-making in healthcare, which will form the basis for future studies and implementation on actual healthcare data.

5.1 Future Work

The developed machine learning models showed much promise in terms of prediction, but there are some potential areas for expanding this work. For the future, larger and more varied datasets from hospitals, health care organisations and electronic health record systems are desirable to enhance the generalisability and robustness of predictive models in populations of diverse sizes. The use of other clinical markers such as blood markers, genetic data, metabolic markers and patient records may also increase prediction accuracy and offer more information regarding obesity risk.

Future research should also consider more sophisticated machine learning and deep learning algorithms such as XGBoost, LightGBM, CatBoost and transformer-based neural networks to see whether greater improvement in predictive performance can be attained. Furthermore, incorporating Explainable Artificial Intelligence methods such as SHAP and LIME would enhance the transparency of the model, enabling healthcare professionals to interpret how different variables contribute to the prediction and promoting greater trust and adoption. A different interesting area could be the development of cloud or mobile healthcare applications that could enable real-time obesity risk assessment and personalised lifestyle suggestions. Finally, machine learning models should be integrated with clinical decision support and external validation with independent datasets to enhance their use in real-world settings and improve clinical adoption

5.2 Reflection

This study gave me an excellent chance to grow technically and to research in the area of data science, healthcare analytics and machine learning. During this project I gained hands-on experience in implementing the entire machine learning pipeline from data processing to exploratory data analysis, feature engineering, model building, hyperparameter tuning, cross-validation and comparative performance analysis. I had the opportunity to work with healthcare data, which deepened my understanding of the practical applications of AI in addressing real-world challenges and promoting evidence-based clinical decisions.

A major difficulty encountered while conducting the research was the choice of a proper machine learning algorithm and hyperparameter tuning for its performance. Even interpreting the measurements of multiclass classification, analysis of confusion matrices and cross-validation results needed a lot of investigation and experimentation. However, the challenges were surmounted with continuous assessment of the model performance and reference to related

academic literature which helped to enhance my analytical and problem-solving ability. Comparative evaluation of different Algorithms further helped in interpreting and realising predictive performance in the light of other than only accuracy as a measure of performance.

Additionally, the project has helped me grow as an academic writer, literature reviewer and critical evaluator, as I was able to integrate theory and practice. Each of the steps taken towards dissertation completion required effective project planning and management of the time allotted to accomplish the task. In an extension of this research, I would include bigger real clinical data sets, more predictive variables and implement more Explainable Artificial Intelligence methods for better model interpretability. This dissertation has given me more confidence to independently research and apply machine learning techniques to solve healthcare problems. The knowledge and hands-on skills that I have gained from this investigation will serve as a solid basis for my future academic research and career in artificial intelligence, healthcare analytics and data science.

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Appendix A: Project Proposal

1. Introduction

1.1 Project overview

Financial technology and cybersecurity are important to credit card fraud detection because monetary losses can result from fraudulent transactions and customers may weaken trust if fraud is repeatedly detected [1]. This project aims at the classification of legitimate payments from fraudulent activities to learn the patterns from the transaction data and enhance the accuracy of classification using supervised machine learning techniques.

1.2 Motivation and rationale

The topic was selected to indicate the growing amount of card transactions due to the increase in digital payments, online banking, mobile wallets and e-commerce. The faster transactions move, the less likely it is that manual fraud review is a viable option. Such automated detection systems are essential for financial institutions to quickly analyse massive amounts of transactions, identify suspicious patterns and create a safer digital payment ecosystem for both customers and the financial institutions themselves [2].

1.3 Dataset and research context

This study focuses on the Kaggle Credit Card Fraud Detection dataset which is a set of anonymised card transaction records from Europe. The most sensitive variables are covertly converted to numeric variables. The class variable is the target variable and the classes are Class = 0 represents genuine transactions and Class = 1 represents fraudulent transactions that need to be detected accurately.

1.4 Project contribution

The project contributes by comparing algorithms of some machine learning models to find the right one for fraud detection. Reliable, balanced and practical evaluation outcomes will be provided by evaluating model performance using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix.

2. Problem Statement

2.1 Main problem

Credit card fraud poses a significant financial and cybersecurity challenge since fraudulent transactions can lead to financial losses, inconvenience to the customer, harm to the bank's reputation and operational pressure for banks and financial institutions [3]. Classical fraud prevention systems are based on fixed rules, manual verification or static transaction thresholds. However fraud techniques continually evolve as criminals figure out how to counter security measures. Therefore rule-based solutions might not be able to detect new or advanced fraudulent patterns particularly in high-volume transactions and in real-time digital banking and online payment systems where instant decisions must be made.

2.2 Who is affected

Several stakeholders are affected by the problem such as cardholders, banks, payment service providers, merchants and financial regulators. Financial stress and card blocking or reduced trust in digital payment could be experienced by customers [4]. Banks and merchants may experience compensation expenses, investigation costs, chargebacks and regulatory pressures to ensure secure financial services and payment providers must guard their infrastructure and effectively prove their compliance with requirements to prevent payment fraud.

2.3 Dataset-specific challenge

The greatest technical difficulty in this project is the extensive imbalance in the credit card fraud dataset. Genuine transactions are much more common than fraudulent transactions making it challenging for models to learn effectively particularly about the minority fraud class. This can lead the user to believe the accuracy is high when in fact a model may classify nearly all transactions as real and obtain a very high accuracy score [5]. Therefore reporting only accuracy could mask sub-optimal fraud detection results and lead to inaccurate conclusions for financial decision-making and for practical deployment scenarios in general.

2.4 Implementation of machine learning

Machine learning is required because it can analyse transaction data, uncover hidden patterns and be more effective than a set of rules in detecting fraud. Supervised classification models can be trained with such relationships between the transaction and the target class and then used for

classifying future transactions as legitimate or fraudulent [6]. Under the right pre-processing, use of imbalance metrics and metrics for assessing the quality of the fraud detection decision, machine learning can help make faster, more adaptive and more reliable fraud detection decisions in large-scale banking systems where manual inspection is slow, costly and inconsistent during real-time operations.

2.5 Research questions

This project will answer four research questions.

RQ1: What transaction patterns help distinguish genuine and fraudulent credit card transactions?

RQ2: How does class imbalance affect fraud detection model performance?

RQ3: Which machine learning model provides the best fraud detection performance?

RQ4: Which evaluation metrics are most suitable for reliable fraud detection?

These questions lead to the study of data understanding, imbalance handling, model comparison and evidence-based evaluation of the reliability of practical fraud detection within financial transaction monitoring systems and security enhancements in banking.

3. Aims and Objectives

3.1 Research aim

To develop and evaluate machine learning models to identify fraudulent credit card transactions using preprocessing, imbalance treatment and comparison.

3.2 Research Objectives

1. To analyse credit card transaction data to identify significant patterns distinguishing genuine transactions from fraudulent financial activities using preprocessing.

2. To examine credit card transaction fraud challenges and factors influencing detection effectiveness across financial systems using machine learning techniques.

3. To develop and compare multiple machine learning classification models for accurately detecting fraudulent credit card transactions using performance metrics.

4. To evaluate classification model performance using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix for reliable fraud detection.

5. To provide recommendations for improving credit card fraud detection systems using optimised machine learning models and future research directions.

3.3 Proposed approach

The practical will start by loading the Kaggle credit card fraud dataset in Python and exploring the structure, data types, descriptive statistics and the distribution of the target variable. Duplicate records and missing values will be identified before the preparation of this dataset for modelling. Since the features V1 to V28 are already anonymised numerical variables, the main scaling focus will be on Time and Amount. The class distribution, transaction amount behaviour, correlations and fraud-related patterns will be explored using exploratory data analysis. The dataset will then be split into a training set and a test set by stratified sampling. The problems of class imbalance will be solved by the SMOTE or class-weighting method which will be used only for training data. Finally several machine learning models will be trained, evaluated and compared with the appropriate evaluation metrics for imbalanced classification and to support valid and reproducible conclusions about the reliability of the models and practical implications in financial monitoring systems.

3.4 Technologies and tools

The primary programming language will be Python, served by the interactive implementation and documentation of Jupyter Notebook [7]. Data processing will be handled by pandas and NumPy and visual analysis will be provided by Matplotlib and Seaborn [8]. Scikit-learn will provide baseline and ensemble classifiers, evaluation metrics and train-test splitting. The imbalanced-learn library will be used to support SMOTE, and XGBoost and LightGBM will be used to enable advanced boosting-based fraud detection models [9]. Overall these tools can be used to preprocess, visualise, model classification, perform reproducible experiments and analyse the performance of this applied research.

4. Legal, Social, Ethical and Professional Considerations

4.1 Legal considerations

The data is publicly available on Kaggle and consists of anonymised credit card transaction data. The project does not include any major privacy risks as cardholder names, account numbers and direct identifiers are not included [10]. However data should be stored securely, only accessed for responsible academic analysis and not used for other purposes.

4.2 Ethical considerations

Ethical issues emerge because of model errors that may impact customers and institutions. If a false positive has been activated, it can be a false alarm and delay payments or prevent them [11]. A false negative can result in genuine fraud going undetected and allow losses to persist. Therefore there is a need to strike a balance between evaluation sensitivity and customer fairness and convenience.

4.3 Social considerations

By preventing financial losses for customers and fostering trust in digital banking, online payments and e-commerce, effective fraud detection can have a positive social impact [12]. Reliable systems can also help reduce stress for victims and enable safer engagement in cashless financial services.

4.4 Professional considerations

Transparency, reproducibility and responsible reporting are expected throughout the project in professional practice. The methodology and the pre-processing steps should be documented clearly so that the results can be reviewed, and the imbalance handling and evaluation process should be documented clearly [13]. Model performance should not be exaggerated particularly for imbalanced fraud data as it only shows the accuracy and not the other metrics. The conclusions and recommendations based on these metrics should be objective using metrics like recall, precision, F1-score and ROC-AUC.

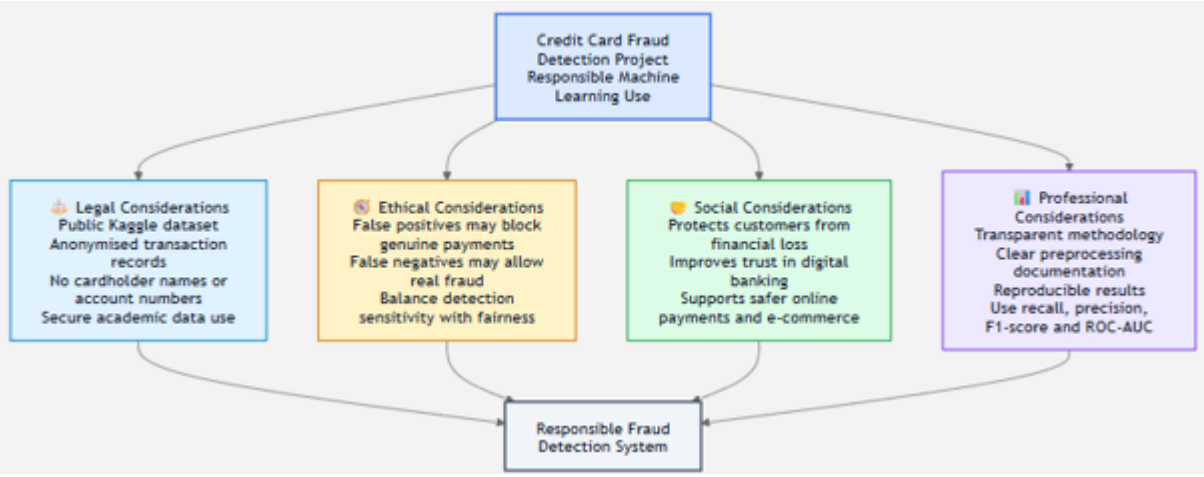


Figure 1: Legal, Social, Ethical and Professional Considerations

(Source: Created by Author, 2026)

5. Background

5.1 Credit card fraud and digital payment risk

The increase in e-commerce, online banking and mobile payment services has made credit card fraud a significant threat to electronic payments. According to the author [14] the digital payment fraud literature has been shifting its attention to e-commerce fraud risk, artificial intelligence, blockchain, information security and fraud prevention. This demonstrates that payment fraud is not just a banking problem but a wider cybersecurity and financial crime problem. Similarly another study [15] states that card fraud can be initiated by the theft of cards, compromise of cardholder data or database breaches which would create significant risk for customers and payment networks. However their study is more about fraud channels and payment challenges while [14] offers a more comprehensive bibliometric analysis of the evolution of research. In contrast [16] highlights a real-world problem that is related to customer confidentiality issues when it comes to real transaction records. Therefore detecting credit card fraud becomes crucial to minimise financial losses, safeguard users and ensure trust in digital transactions.

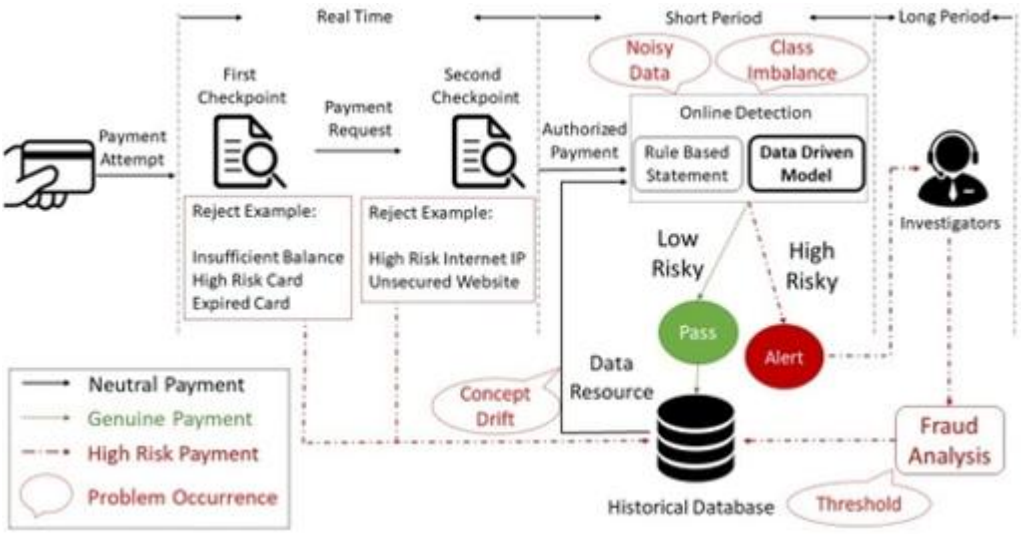


Figure 2: Credit card fraud and digital payment risk

(Source: [17])

5.2 Fraud Detection using Machine Learning

Fraud detection uses machine learning to identify fraudulent transactions by analysing transactions to detect patterns and then categorising future transactions based on those patterns. According to the author [18] traditional fraud detection methods relying on rules are charged with disadvantages as they are rigid, generate high false positives and have limitations in processing large amounts of financial data in real-time. They believe machine learning can provide more adaptive solutions through predictive analytics, anomaly detection and deep learning. Similarly another study [19] indicate that there is a rising use of AI and machine learning in the field of financial crime detection which is used to uncover intricate patterns of financial crime and enhance financial monitoring. However the author [19] also argues that the field of fraud analytics with machine learning is still in development. Supervised classification is appropriate for this project because there are genuine and fraudulent transactions present in the data set and they are labelled. Unlike unsupervised anomaly detection, supervised models can be trained under the guidance of fraud labels. However machine learning is not an automatic solver for all fraud detection challenges as imbalanced data, regulatory concerns, computational expense and model interpretability are all significant problems.

5.3 Challenges of imbalanced fraud datasets

One of the most significant difficulties in credit card fraud detection is the unbalanced nature of the data sets where fraudulent transactions are far fewer in number than legitimate transactions. The author [20] states that biased predictions, poor generalisation and high false negative rates can result from imbalanced fraud datasets. It is dangerous because actual fraud cannot be caught by the model and it might seem successful. Therefore accuracy is not sufficient most transactions can be predicted to be correct by a model and still be extremely accurate.



Figure 3: Challenges of imbalanced fraud datasets

(Source: [21])

Similarly another author [22] demonstrates that the sampling methods yield different trade-offs. Random under-sampling may give a more precise result but can also lead to a decrease in recall while SMOTE can give a more balanced F1 score. In contrast another study [23] suggest that as their embedding-aware conditional GAN produced high-quality minority-class data the use of advanced synthetic generation has the potential to outperform traditional oversampling techniques, which is the case for LightGBM. In contrast, another author's study states [24] that are concerned with feature selection rather than sample generation and proposes FID-SOM for highly imbalanced data. These studies concur that imbalance is a problem to be solved but differ on the best remedy which is either resampling, synthetic generation or feature selection. In this project one can use SMOTE or class weighting but resampling should be performed on the training data only to prevent data leakage.

5.4 Data Preprocessing and Feature Preparation

Data preprocessing is crucial as it ensures that the models for fraud detection are fed with clean and well-prepared transaction data. According to the study [25], effective fraud detection relies on preprocessing, resampling and model optimisation and the authors applied SMOTE and hyperparameter tuning to enhance the predictive accuracy. Similarly another author [26] has exploratory data analysis, duplicate data checking, outlier detection, feature scaling, encoding and data balancing before modelling. Preprocessing is not only a technical process but a significant component of model reliability. However the study [27] focuses more on a systematic approach with several different data sets such as missing value imputation and dimensionality reduction to enhance the generalisability of the results. For this project the missing values and duplicated values will be checked before modelling. The main scaling requirement is that of Time and Amount in this case since V1–V28 are all numerical features and already anonymised. A split strategy of the train set into two strata will also maintain the same fraud rate in both sets of data.

5.5 Models for Fraud using Machine Learning Techniques

Logistic Regression

Logistic Regression is an easy-to-interpret baseline classification model for binary fraud classification with clear estimates and probabilities for genuine or fraudulent transactions. According to the author [28] strong scaling, under-sampling, cross-validation, and PR-AUC evaluations were seen to help logistic regression in its effective performance. However the results of the author [29] indicated that Random Forest outperformed all the other models in this study which indicates that logistic regression serves as a valuable baseline model but might not be optimal when dealing with more complex and non-linear patterns in fraud.

Decision Tree

Decision Tree models are easy to interpret working by dividing the transactions into different categories based on the features. The author states that [30] the decision-tree algorithms can learn and recognise patterns and deviations from normal behaviour which can assist in automated card fraud detection. In contrast another author [31] evaluate decision trees, which can overfit in a broader explainable AI landscape and finds that tree models are useful but there is the potential for better-performing ensemble models.

Random Forest

Random Forest is based on the principle of using several trees to enhance a single tree typically resulting in greater stability and less overfitting. According to the study [29] Random Forest had the best performance with high accuracy and AUC compared to Logistic Regression, Decision Tree and Random Forest. Similarly another study [32] has demonstrated that Random Forest can eliminate false alarms using high precision in digital payment fraud detection. However the rate they recalled was lower and some cases of fraud may be missed. Therefore Random Forest is strong but needs to be evaluated based on Recall and F1 score and not just Precision.

XGBoost and LightGBM

The boosting models include XGBoost and LightGBM which can build sequential learners to correct earlier mistakes and are appropriate for structured fraud sets. The author [33] indicates that models like CatBoost, XGBoost and LightGBM are useful for fraud detection but CatBoost has the highest F1 score in their comparison. Similarly another author [34] demonstrates that the XGBoost model can play a significant role when part of an ensemble consisting of several models, the ensemble enhances the F1-score and the precision-recall trade-off. However booster models are more complex and require more tuning and computational power than simpler models.

Model comparison justification

The comparison between Logistic Regression, Decision Tree, Random Forest, XGBoost and LightGBM offers a comprehensive evaluation of fraud detection models. Logistic Regression is an approach that gives interpretability which is recommended by [28] and Random Forest is recommended by the author [29]. However the study of the author [33] demonstrates that models that are boosted do very well on high-dimensional transactional data.

5.6 Evaluation Metrics

With fewer fraudulent transactions, performance in fraud detection is measured in other metrics besides accuracy. According to the author [35] the accuracy, F1-score and the confusion matrix are helpful for credit card fraud assessment. However the study [22] demonstrates that sampling methods cause compromises between recall and precision. Recall is used to find fraud cases that are missed, and precision is used to reduce the number of false alarms. ROC-AUC provides a measure of separation between the true positive and false positive rates and the confusion matrix

provides a practical interpretation of false positives and false negatives in financial transaction monitoring systems in general contexts.

5.7 Research gap and summary

The literature displays significant advances in fraud detection but these advances are accompanied by high accuracy in detecting fraud which has not been properly addressed considering the severe class imbalance. The author [20] states that the imbalance leads to a higher proportion of false negatives. However in addition to accuracy the practical reliability demands models to be compared through other measures like Recall, Precision, F1-score or ROC-AUC in deployment settings. Detecting credit card fraud is a well-known machine learning problem but still difficult due to the dynamic nature of fraud patterns and the rarity of fraudulent transactions. Similarly today the literature has established that the need for preprocessing, dealing with imbalance and having a correct evaluation method and metrics of the models is essential to build practical and trustworthy systems for fraud detection.

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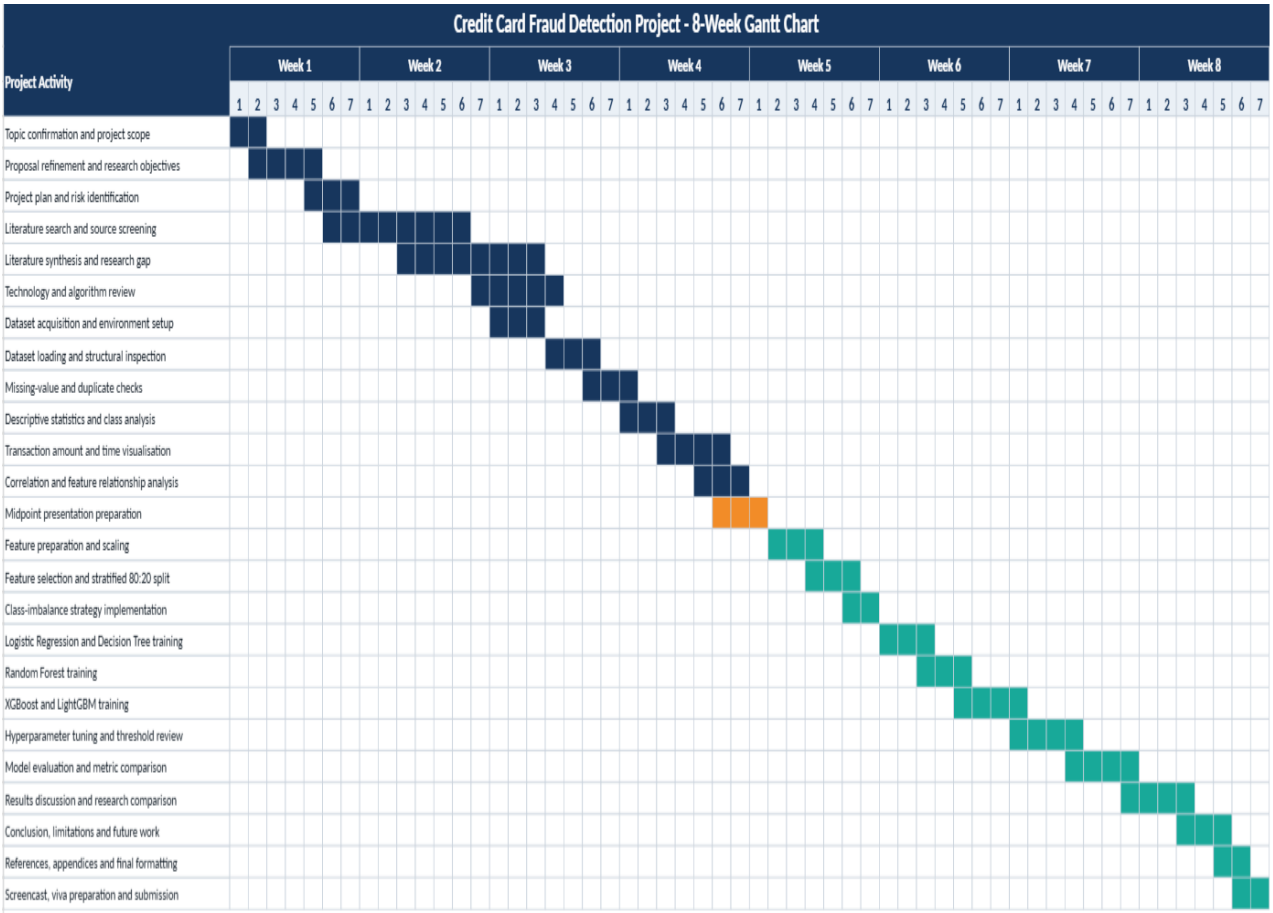
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Appendix B: Project Management



Appendix C: Artefact/Dataset

Dataset Link: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Appendix D: Screencast

Github Link: https://github.com/DEVADHARSANSELVENDRAN/FINAL-REPORT